

Notes on superpatterns: an improved lower bound, short superpatterns for $k = 7, 8$, and exact values for small alphabets and rosaries

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Abstract

A permutation $\sigma \in S_n$ is a k -superpattern if it contains every $\pi \in S_k$ as a pattern. Let $\text{sp}(k)$ be the least such n . We prove an explicit finite inequality for the number of distinct k -patterns of any permutation of length n (Theorem A), and deduce $\text{sp}(k) \geq (1.0003125 - o(1)) k^2/e^2$ with a proof shorter than that of Chroman, Kwan and Singhal (2021), whose constant is 1.000076; a refinement that counts value slots instead of positions (Theorem C) gives $\text{sp}(k) \geq (1.00734 - o(1)) k^2/e^2$, multiplying the excess over the trivial bound k^2/e^2 by 96. We also give explicit 7- and 8-superpatterns of lengths 23 and 30 (previous best 25 and 33), determine the exact minimal length of k -superpatterns over the alphabet $[k+1]$ for $k \leq 5$ (answering Problem 3 of Hunter for these k), and verify Gupta’s rosary conjecture $r(n) \leq \lfloor n^2/2 \rfloor$ for all $n \leq 11$. All computational claims come with certificates that can be re-checked in seconds, and Theorem A, its numerical constant, and the certificates for $k = 7, 8$ are additionally verified in Lean 4.

1 Introduction and summary of results

Let $\sigma \in S_n$ and $\pi \in S_k$. We say σ *contains* π (as a pattern) if there are indices $t_1 < \dots < t_k$ with $\sigma(t_i) < \sigma(t_j) \iff \pi(i) < \pi(j)$. A permutation containing every $\pi \in S_k$ is a *k -superpattern* (Arratia [1]; “ k -universal” in [2]). Write $\text{sp}(k)$ for the minimum length of a k -superpattern (OEIS A342474) and $\text{pat}_k(\sigma)$ for the number of distinct k -patterns contained in σ .

Previously known.

- $\text{sp}(k) = 1, 3, 5, 9, 13, 17$ for $k = 1, \dots, 6$ [3].
- Upper bound $\text{sp}(k) \leq \lceil (k^2 + 1)/2 \rceil$ [3, Thm 6.4], improving $(k^2 + k)/2$ [4] and k^2 [1].
- Lower bound $\text{sp}(k) \geq (1.000076/e^2) k^2$ for large k [2, Thm 1.1], improving the trivial $\binom{n}{k} \geq k!$ bound $(1/e^2 - o(1))k^2$ [1] and disproving Arratia’s conjecture that k^2/e^2 is tight.
- For words over the alphabet $[k+1]$, Miller’s construction has length $(k^2 + k)/2$; Hunter [5, Thm 1] proved this is asymptotically optimal and asked (Problem 3) whether $f(k; k+1) = (k^2 + k)/2$ exactly.
- Gupta’s rosary conjecture $r(n) \leq n^2/2$ is known for even n [6]; for odd n only $r(n) < n^2/2 + n/4 - 1$.

Results of this note.

Table 1: Results of this note, with the previously best known value and how each is verified.

Result	Previous best	This note	Verified by
Lower bound $\text{sp}(k) \geq \lambda k^2/e^2$ (Thm A, §2)	$\lambda = 1.000076$ [2]	$\lambda = 1.0003125$ (excess $\times 4.1$); shorter proof	proof; numerics reproduced twice; Theorem A and the constant fully verified in Lean 4
Lower bound, slot refinement (Thm C, §2)	$\lambda = 1.000076$ [2]	$\lambda = 1.00734$ (excess $\times 96$)	proof; per-class lemma brute-forced; rate reproduced twice; finite- k certificates
$\text{sp}(7)$ (Thm 2, §3)	≤ 25	≤ 23	certificate; brute force; Lean <code>native_decide</code>
$\text{sp}(8)$ (Thm 2, §3)	≤ 33	≤ 30	certificate; brute force; Lean <code>native_decide</code>
Structured 7-superpattern; Arnason's 17 explained (§4)	25; unexplained	length 24; tie-broken 3-block word	two checkers
$f(k; k+1)$, $k \leq 5$ (Thm 3, §5)	$\leq (k^2 + k)/2$	$= (k^2 + k)/2$ (Hunter's Problem 3)	SAT unsatisfiability + witness check
$f(k; k+2)$, $k = 3, 4$ (§5) Gupta rosaries $r(n)$ (Thm 4, §6)	— odd n : $< n^2/2 + n/4 - 1$ [6]	$= \text{sp}(k)$ $= \lfloor n^2/2 \rfloor$ for $n \leq 5$; $\leq \lfloor n^2/2 \rfloor$ for $n \leq 11$	SAT SAT / explicit rosaries + checker
$t(9), t(10)$ (§7) $0.787 \leq c_{21} \leq 1.140$; every $\pi \in S_k$ absent from Π_N w.h.p. if $k > 2.279\sqrt{N}$ (§7) Every union of $\leq r$ runs contained at $n = O(r^3 \log r k^2)$ with failure $e^{-\Omega(n/r^3)}$ (Thm 9, §7)	unknown $0.40 \leq c_{21} \leq e/2$; $k > e\sqrt{N}$ threads: failure only $e^{-O(k)}$	$\approx 60, \approx 70\text{--}72$ threshold of every pattern $\geq 0.1925k^2$ speed N	sampling proof; certificate recomputed
Every block-grid pattern (tilted grids, the family $\mathcal{F}$, $k = rh$) contained in a random permutation of length $(\pi/8 + o(1))k^2$ when $\min(r, h) \rightarrow \infty$ (Thm 10, §7)	$k^2 \text{ polylog } k$ (threads; He-Kwan)	r -independent threshold $0.393k^2$, within $\pi/2$ of the conjectured $k^2/4$	proof; greedy simulated with copy check
Every $\pi \in S_k$ contained in a random permutation of length $0.757k^2$ w.h.p., failure $e^{-\eta k}$ (Thm 11, §7)	k^2 (rigid rows = He-Kwan threads); $20k^2$ for quasirandom π [7]	first uniform per-pattern threshold below k^2	reduction proved; constant from quadrature, re-simulated

Result	Previous best	This note	Verified by
Identity is the <i>hardest</i> pattern: $n_{1/2}/k^2$ for random π is $0.881 \times$ the identity's at $k = 40$ and decreasing; random-pattern limit $\approx 0.22 \pm 0.02 < 1/4$ (§7)	unknown; folklore “identity easiest”	the constant $1/4$ is the maximum over patterns; $\kappa_{\text{univ}} > 2$	numerics (600–4000 samples/pt, $k \leq 40$), solver cross-checked
Speed- N tail for (12) ^h : $\Pr \leq e^{-\rho(C)N}$, $\rho > 0$ for $C > 27.6$ (Thm 13, §7)	speed $N/12$ at threshold ≈ 51 (box certificates)	first k -uniform speed- N bound beyond monotone/run patterns	proof (alternating-chain certificate; constant recomputed)
Speed- N tails, uniform in k , for direct sums of bounded blocks (e.g. $(21)^{k/2}$: $e^{-0.122N}$ for $N \geq 24k^2$) and for all block-grid patterns $\mathcal{G}(r, h)$ (rate $\Theta(N/r^2 \ln r)$ for $N \geq 16(2 \ln r + 2)k^2$) (Thm 14, §7)	greedy: failure speed $\leq \min(k, h)$; periodic word only, threshold $9r \ln(3er)k^2$	local (cell-count) mechanism, no LIS structure; grid barrier isolated as chain-vs-lattice	proof (Mirsky/negative association; constants recomputed)
Union-bound slack $R = \mathbb{E}M / \Pr(M > 0)$ at threshold: $\ln R \approx 0.65k$ ($k = 5\text{--}9$), $\rightarrow 0$ by $1.1k^2$; missing patterns cluster under adjacent transpositions (§7)	Janson, Bonferroni void	first moment off by $e^{\Theta(k)}$ at $t(k)$; witness reduction $R \leq \max_{\pi} \mathbb{E}[M \mid \pi \not\subseteq \sigma_n]$ (Prop. 15) makes the threshold constant a first-moment constant if $R = e^{o(k^2)}$	exhaustive counts; $k = 6$ re-sampled independently; proofs checked
A uniformly random $\pi \in S_k$ is contained in a random permutation of length $0.527k^2$ (Thm 16) and of length $0.465k^2$ (Thm 17, in-gap lookahead with certified Bellman constant) w.h.p.; value-blind gap rules capped at $\frac{1}{2}$, all in-gap rules at ≈ 0.462 (§7)	$0.757k^2$ for every π (Thm 11); $20k^2$ for quasirandom π [7]	first separation of typical from worst-case pattern; first constant below $\frac{1}{2}$	reductions proved; constants recomputed independently
Block-grid patterns (tilted grids) contained at $0.266k^2$ via cross-strip lookahead (blocks of 64); full lookahead numerically $\frac{1}{4}$ exactly (Thm 18, §7)	$(\pi/8)k^2 = 0.393k^2$ (Thm 10); k^2 polylog (threads)	first constant below $\pi/8$; identifies cross-strip lookahead as the barrier's content	reduction proved; constants recomputed independently (MC)

Result	Previous best	This note	Verified by
Every tilted grid $(12 \cdots r)^h$ (and every block-grid pattern with bounded strip revisits), $\min(r, h) \geq (\ln k)^3$, contained at $(\frac{1}{4} + \varepsilon)k^2$ for every $\varepsilon > 0$; $\gamma_\infty = 1$ with $ \mathbb{E}K_n - n \leq \sqrt{2n} + \frac{1}{2}$ (Thm 20, §7)	$0.266k^2$ (Thm 18); $(\pi/8)k^2$ (Thm 10)	first non-monotone family at Alon's constant; Burke property of the cross-strip recursion; block rules capped at $\frac{1}{4}$	proof; Burke law, $\mathbb{E}K_n$ and certified constants recomputed independently
Fixed-strip constant of tilted grids: $\limsup_r C^{\text{fix}}(r, h) \leq C_{\text{mf}}(h)$, $C_{\text{mf}}(h) \downarrow \pi/8$; the pattern $(12)^h$ is numerically <i>easier</i> than the identity (Thm 12, §7)	—	first exact-threshold theory for a non-monotone family; evidence that $1/4$ is not the minimum over patterns	proof; Bellman constant recomputed
Random perm. of length $72k^2$ contains all π with $L_\Delta(\pi) \leq k/3 \ln^2 k$ for all but $k/3 \ln k$ shifts (Thm 8, §7)	$20k^2$ for $L_\Delta \leq 3\sqrt{k}$ all Δ [7]	exceptional set $k!e^{-\Theta(k/\ln k)}$ vs $k!e^{-\Theta(\sqrt{k})}$	proof; overlap lemma simulated

Result 1 is the main theorem. Results 2–6 are computational; their value is that each is a concrete data point with an easily re-checkable certificate, and together they extend the known tables (A342474 stops at $k = 6$).

2 The lower bound

Throughout, k is odd (the even case follows since $\text{sp}(k+1) \geq \text{sp}(k)$). A k -subset $T = \{t_1 < \cdots < t_k\} \subseteq [n]$ corresponds bijectively to the *gap vector* $a = (a_0, \dots, a_k)$, $a_0 = t_1$, $a_j = t_{j+1} - t_j$ ($1 \leq j < k$), $a_k = n + 1 - t_k$: a composition of $n + 1$ into $k + 1$ positive parts. The *width* of index i ($2 \leq i \leq k - 1$) is $b_i = t_{i+1} - t_{i-1} = a_{i-1} + a_i$.

2.1 Stable encodings

The starting point is the encoding idea of [2]: a pattern π occurring at T can be recorded by giving the positions t_j for $j \notin I$ and the *values* $\pi(i)$ for $i \in I$, which is cheaper than giving t_i whenever the window (t_{i-1}, t_{i+1}) has more than k integer points. CKS fix a threshold and a set of exactly ck indices and pay a large-deviation estimate for subsets with few wide windows. The following lemma removes the threshold and the case distinction.

Let $\mathcal{I}$ be a map $T \mapsto I(T) \subseteq \{2, \dots, k-1\}$ such that $I(T)$ contains no two consecutive integers and is *stable*: whenever T' agrees with T outside $I(T)$ (i.e. $t'_j = t_j$ for all $j \notin I(T)$), we have $I(T') = I(T)$.

Lemma 1. For every stable $\mathcal{J}$ and every $\sigma \in S_n$,

$$\text{pat}_k(\sigma) \leq \sum_{T \in \binom{[n]}{k}} \frac{k!}{(k - |I(T)|)!} \prod_{i \in I(T)} \frac{1}{b_i(T) - 1}.$$

Proof. For each π contained in σ fix one occurrence $T = T(\pi)$ and set $I = I(T)$, $\varphi(\pi) = (I, (t_j)_{j \notin I}, (\pi(i))_{i \in I})$. φ is injective: from σ and $(t_j)_{j \notin I}$ we know the relative order of the values $\sigma(t_j)$, $j \notin I$; from $(\pi(i))_{i \in I}$ we know which $k - |I|$ ranks in $[k]$ are taken by the indices $j \notin I$; together these determine π . Group the codes by $\psi = (I, (t_j)_{j \notin I})$; each ψ admits at most $k!/(k - |I|)!$ third components. Let $\text{ext}(\psi)$ be the set of $T' \in \binom{[n]}{k}$ agreeing with ψ off I . Since I has no two consecutive elements, t'_i ranges independently over the open interval (t_{i-1}, t_{i+1}) for $i \in I$, so $|\text{ext}(\psi)| = \prod_{i \in I} (b_i - 1)$, where b_i is determined by ψ . By stability, every $T' \in \text{ext}(\psi)$ has $I(T') = I$, so T' determines ψ and the sets $\text{ext}(\psi)$ are pairwise disjoint. Hence

$$\begin{aligned} \sum_{\psi} \frac{k!}{(k - |I|)!} &= \sum_{\psi} \sum_{T' \in \text{ext}(\psi)} \frac{k!}{(k - |I|)! |\text{ext}(\psi)|} \\ &\leq \sum_{T'} \frac{k!}{(k - |I(T')|)!} \prod_{i \in I(T')} \frac{1}{b_i(T') - 1}. \quad \blacksquare \end{aligned}$$

2.2 The even rule and the tilt

Take $I(T) = \{i \text{ even} : b_i - 1 > k\}$. It is non-adjacent, and stable because it depends only on the odd-indexed positions $t_{i \pm 1}$, none of which lie in $I(T)$. Using $k!/(k - |I|)! \leq k^{|I|}$, Lemma 1 gives

$$\text{pat}_k(\sigma) \leq \sum_T W(T), \quad W(T) = \prod_{i \text{ even}} f(b_i), \quad f(b) = \min\left(1, \frac{k}{b - 1}\right). \quad (1)$$

In gap coordinates, $W(a) = \prod_{j=1}^{(k-1)/2} f(a_{2j-1} + a_{2j})$: a product over *disjoint* pairs of gaps. For any $x \in (0, 1)$, since $\sum a = n + 1$ on every term,

$$\begin{aligned} \sum_T W(T) &= x^{-(n+1)} \sum_{a: \sum a = n+1} x^{\sum a} W(a) \leq x^{-(n+1)} \sum_{a \in \mathbb{Z}_{\geq 1}^{k+1}} x^{\sum a} W(a) \\ &= x^{-(n+1)} \left(\frac{x}{1-x}\right)^2 \prod_{j=1}^{(k-1)/2} \sum_{a, a' \geq 1} x^{a+a'} f(a + a'). \end{aligned} \quad (2)$$

(The free gaps a_0, a_k contribute $\sum_{a \geq 1} x^a = x/(1-x)$ each.)

Lemma 2. For $x \in (0, 1)$ and $k \geq 1$, $\sum_{a, a' \geq 1} x^{a+a'} f(a + a') = \left(\frac{x}{1-x}\right)^2 (1 - x^k)$.

Proof. Writing $s = a + a'$, there are $s - 1$ pairs with sum s . Split at $s = k + 1$:

$$\sum_{s=2}^{k+1} (s-1)x^s + \sum_{s \geq k+2} (s-1)x^s \frac{k}{s-1} = \left[\sum_{s \geq 2} (s-1)x^s - \sum_{s \geq k+2} (s-1)x^s \right] + \frac{k x^{k+2}}{1-x}.$$

Now $\sum_{s \geq 2} (s-1)x^s = x^2/(1-x)^2$ and $\sum_{s \geq k+2} (s-1)x^s = x^{k+2}[(k+1)/(1-x) + x/(1-x)^2]$. Collecting terms over the common denominator $(1-x)^2$:

$$\frac{x^2 - x^{k+2}((k+1)(1-x) + x) + kx^{k+2}(1-x)}{(1-x)^2} = \frac{x^2 - x^{k+2}}{(1-x)^2} = \left(\frac{x}{1-x}\right)^2 (1 - x^k).$$

■

Combining (1), (2) and Lemma 2:

Theorem A. For every $\sigma \in S_n$, every odd $k \geq 3$ and every $x \in (0, 1)$,

$$\text{pat}_k(\sigma) \leq x^{-(n+1)} \left(\frac{x}{1-x} \right)^{k+1} (1-x^k)^{(k-1)/2}. \quad (\text{A})$$

Note that dropping the factor $(1-x^k)^{(k-1)/2}$ and optimizing x recovers exactly the trivial bound $\binom{n}{k}$ up to lower-order terms; the whole gain is in that factor.

2.3 Asymptotics

Put $n = \lambda k^2/e^2$ and $x = e^{-\theta/k}$ with $\theta > 0$ fixed. Then $-(n+1)\log x = (n+1)\theta/k$, $\log \frac{x}{1-x} = \log \frac{k}{\theta} - \frac{\theta}{2k} + O(k^{-2})$, and $\log k! = k \log k - k + O(\log k)$; hence

$$\frac{1}{k} \log \frac{\text{RHS(A)}}{k!} = \underbrace{\frac{\lambda\theta}{e^2} - \log \theta + 1 + \frac{1}{2} \log(1 - e^{-\theta})}_{=: g(\theta, \lambda)} + O\left(\frac{\log k}{k}\right). \quad (\text{A}')$$

$g(\cdot, \lambda)$ is strictly convex on $(0, \infty)$ (it is $\lambda\theta/e^2 - \log \theta$ plus a convex function), with a unique minimizer $\theta^*(\lambda)$ solving $\lambda/e^2 = 1/\theta - \frac{1}{2}e^{-\theta}/(1 - e^{-\theta})$, and $\min_{\theta} g(\theta, \lambda)$ is continuous and strictly increasing in λ . Numerically (mpmath, 30 digits, two independent implementations):

$$\begin{aligned} \theta^*(1) &= 7.37192\dots, & g(\theta^*(1), 1) &= -3.1174 \times 10^{-4}, \\ \lambda_A &:= \text{root of } \min_{\theta} g(\theta, \lambda) &= 1.00031251\dots, & \theta^*(\lambda_A) &= 7.36959\dots \end{aligned}$$

Hence for every fixed $\lambda < \lambda_A$ and all large k , every $\sigma \in S_n$ with $n \leq \lambda k^2/e^2$ satisfies $\text{pat}_k(\sigma) \leq e^{-ck} k!$ for some $c = c(\lambda) > 0$; in particular σ is not a k -superpattern.

Corollary. $\text{sp}(k) \geq (1.0003125 - o(1)) k^2/e^2$.

For a fully explicit finite check, at $k = 100001$, $\lambda = 1.00031$, $\theta = \theta^*(\lambda)$, the exact right-hand side of (A) gives $\frac{1}{k} \log(\text{RHS}/k!) = -1.09 \times 10^{-5} < 0$.

Comparison with [2]. Their encoding specifies values for exactly ck even indices ($c = 0.00075$) whose widths exceed dn/k ($d = 8.180$), and needs their Lemma 2.1 (a Chernoff bound showing that all but $O(\lambda^k) \binom{n}{k}$ subsets have that many wide windows); this yields $\log \lambda_{\text{CKS}} \approx c \log(d/e^2) \approx 7.6 \times 10^{-5}$. In Theorem A the right-hand side of Lemma 1 is an *expectation of a product*, so after the exponential tilt every wide window contributes its full saving $\min(1, \theta/B)$, $B \sim \Gamma(2, 1)$ in the limit, with $\mathbb{E} \min(1, \theta/B) = 1 - e^{-\theta}$; no case split is needed. The tilt itself costs only $\tau - 1 - \log \tau \approx 2.7 \times 10^{-6}$ where $\tau = \theta\lambda/e^2 \approx 0.9977$.

We stress that CKS anticipated this: their concluding remarks note that “with a more sophisticated argument one could take both even and odd i into account” and that “presumably one could prove exact large deviation bounds”, but that fully optimizing “would be very complicated”. Lemma 1 is what makes the optimization short: once the bound is written as a sum of products over subsets, the exponential tilt does the large-deviation work exactly and for free. The novelty is therefore in the formulation, not in a new mechanism; a priority search (citations of [2], [5], [3]; arXiv full text; OEIS A342474; Bóna [8]) found no improvement of the constant 1.000076 in the literature through August 2026.

2.4 The value-slot refinement: Theorem C

Lemma 1 charges index $i \in I$ the factor $1/(b_i - 1)$ because t_i has $b_i - 1$ possible positions, each of which could in principle give a different pattern. But the pattern only depends on where $\sigma(t_i)$ falls among the other chosen values, so the number of distinct patterns obtainable by moving t_i is at most the number of *value slots* actually hit by σ on the window. For T with $I(T) = \{\iota_1 < \dots < \iota_r\}$ let $V(T) = \sigma(\{t_j : j \notin I(T)\})$ and let $M_i(T)$ be the number of distinct slots relative to $V(T)$ (a slot of u is $\#\{v \in V : v < u\}$) hit by the values $\sigma(p)$, $p \in (t_{i-1}, t_{i+1})$.

Lemma 1 . For every stable rule $\mathcal{J}$,

$$\text{pat}_k(\sigma) \leq \sum_T \min\left(1, \prod_{j=1}^{r(T)} \frac{\min(M_{\iota_j}(T) + j - 1, k)}{b_{\iota_j}(T) - 1}\right).$$

Proof. As in Lemma 1, group witnesses into classes $\text{ext}(\psi)$; the number of distinct patterns in a class is trivially at most $|\text{ext}(\psi)| = \prod (b_i - 1)$. For the other bound, insert the I -values $v_j = \sigma(t'_{\iota_j})$ one at a time into the sorted list of V and let s_j be the slot of v_j relative to $V_j = V \cup \{v_1, \dots, v_{j-1}\}$; the sequence $(s_1, \dots, s_r)$ determines the pattern. A prefix $(s_1, \dots, s_{j-1})$ determines the V -slot c_l of each earlier v_l and their relative order within each V -slot; for a window value u in V -slot c one has $\text{slot}(u | V_j) = c + \#\{l < j : c_l < c\} + \#\{l < j : c_l = c, v_l < u\}$, and the last term lies in $[0, n_c]$ with $n_c = \#\{l < j : c_l = c\}$. Hence, over *all* realizations of the prefix, s_j takes at most $\sum_{c \text{ hit}} (1 + n_c) \leq M_{\iota_j} + j - 1$ values, and also at most $|V_j| + 1 \leq k$. Multiply over j . ■

(The per-factor cap $b_{\iota_j} - 1$ would be false here: realizations of the same prefix with different values v_l send a fixed window position to different slots. We caught this in a first draft by enumerating every class on the $\text{sp}(6) = 17$ permutation; the corrected bound has no violations on all permutations tested.)

The slot counts are controlled uniformly in σ by a domination argument. Under the geometric tilt, T is the set of the first k points of a Bernoulli($1-x$) site process B on $[n]$; the off- I values are contained in $\sigma(B \setminus [t_{i-1}, t_{i+1}]) \cup \{\sigma(t_{i-1}), \sigma(t_{i+1})\}$, and the number of slots hit relative to $\sigma(B \setminus [t_{i-1}, t_{i+1}])$ equals $1 + \sum_{\ell} \mathbf{1}[B \cap A_{\ell} \neq \emptyset]$ where the A_{ℓ} (positions of the values lying strictly between consecutive window values) are *disjoint* sets outside the window. These indicators are independent, independent of B inside the window, and by Jensen have total mean at most $m(b) = (b-2)(1-x^{(n-b-1)/(b-2)})$. A Chernoff bound per window and a union bound over the $O(kn)$ windows then give, for every $\sigma \in S_n$, $x \in (0, 1)$, threshold β and $\varepsilon > 0$:

Theorem C. With $q = 1-x$, $h(\varepsilon) = (1+\varepsilon)\log(1+\varepsilon) - \varepsilon$, $P(b) = (b-1)q^2x^{b-2}$ and $e = \lfloor (k-1)/2 \rfloor$ i.i.d. even widths $b_1, \dots, b_e \sim P$, $I = \{\iota : b_{\iota} \geq \beta\} = \{\iota_1 < \dots < \iota_r\}$,

$$\begin{aligned} \text{pat}_k(\sigma) \leq x^{-(n+1)} \left(\frac{x}{1-x}\right)^{k+1} & \left(\mathbb{E} \min\left(1, \prod_{j=1}^r \frac{\min((1+\varepsilon)m(b_{\iota_j}) + j + 2, k)}{b_{\iota_j} - 1}\right) \right. \\ & \left. + e n q \sum_{b \geq \beta} P(b) e^{-m(b)h(\varepsilon)} \right). \end{aligned}$$

The full proof (with the $+j+2$ bookkeeping and the Bernoulli representation) is in `experiments/w7-slots/proof.md`. Passing to the limit as in Theorem A, with $b = (k/\theta)B$, $B \sim \Gamma(2, 1)$, $\beta = \bar{\beta}k$, the exponential rate is

$$\tau - 1 - \log \tau + \log \lambda + \max(R_{\text{good}}, R_{\text{bad}}), \quad R_{\text{bad}} = -\bar{\beta}(1 - e^{-\tau/\bar{\beta}})h(\varepsilon),$$

$$R_{\text{good}} = \max_{s \geq 0} \left[-\frac{\bar{c}s^2}{2} + \frac{1}{2} \log \left(\Pr(B < \theta\bar{\beta}) + e^s(1+\varepsilon) \mathbb{E}[(1-e^{-\tau\theta/B}); B \geq \theta\bar{\beta}] \right) \right], \quad \bar{c} = (1+\varepsilon)\bar{\beta}(1-e^{-\tau/\bar{\beta}}),$$

where the s -maximization accounts for the $+j$ terms. Optimizing ($\tau = 0.977$, $\bar{\beta} = 0.594$ — i.e. one drops even indices whose window exceeds $0.59k$, far more aggressive than Theorem A's $b > k + 1$ — and $\varepsilon = 0.149$) the root is

$$\lambda_C = 1.00483,$$

computed independently twice (the second implementation brackets the root in $(1.0048, 1.005)$).

Corollary. $\text{sp}(k) \geq (1.00483 - o(1)) k^2/e^2$.

Sharpening the bad term (Theorem C'). The Chernoff bound $e^{-mh(\varepsilon)}$ is the Poisson-regime bound, but the slot count of a window is a sum of $b - 2$ independent indicators with individual means $1 - x^{|A_\ell|}$ that are *not* small (about 0.6–0.85 at the optimum), where the exact Bernoulli (Cramér) tail is much sharper. Since $f(s) = \log(1 + (e^\lambda - 1)(1 - x^s))$ is increasing and concave in s , Jensen with $\sum_\ell |A_\ell| \leq n - b - 1$ shows that the Laplace transform of the slot count is dominated by that of $\text{Bin}(b - 2, \bar{p}_b)$ with $\bar{p}_b = m(b)/(b - 2)$, hence

$$\Pr(S \geq (1 + \varepsilon)m(b)) \leq \exp(-(b - 2) D((1 + \varepsilon)\bar{p}_b \| \bar{p}_b)), \quad D(a \| p) = a \log \frac{a}{p} + (1 - a) \log \frac{1 - a}{1 - p},$$

(and $\Pr = 0$ if $(1 + \varepsilon)\bar{p}_b > 1$). Replacing $e^{-m(b)h(\varepsilon)}$ by this in Theorem C, and choosing a width-dependent ε_b so that the exponent equals a constant Rk for every width (which removes the need for monotonicity of the bad rate in b), the limit rate becomes $\tau - 1 - \log \tau + \log \lambda + \max(R_{\text{good}}, -R)$ with the same R_{good} (now with $(1 + \varepsilon_b)$ inside the expectation and $\bar{c} = \min_b (1 + \varepsilon_b)m(b)/k$). Optimizing ($\tau = 0.968$, $\beta = 0.537$, $R = 0.00783$, slack $\varepsilon_\beta = 0.072$ at the narrowest kept window) gives

$$\lambda_{C'} = 1.00734,$$

the root being reproduced by two independent implementations, and the finite- k inequality evaluated exactly in log-space (all truncation tails added: widths beyond $4000k/\theta$, $r > 0.14k$) gives $\frac{1}{k} \log(\text{RHS}/k!) = -1.4 \times 10^{-4}$ at $k = 10^5$ and -2.6×10^{-4} at $k = 3 \cdot 10^5$ for $\lambda = 1.007$ (limit value -3.3×10^{-4} ; [experiments/w25-asymptopia-review/finite_check_bern.py](#)).

Corollary'. $\text{sp}(k) \geq (1.00734 - o(1)) k^2/e^2$: the excess over k^2/e^2 is 96 times that of [2] and 23 times that of Theorem A.

For Theorem C itself the excess is fifteen times that of Theorem A and 63 times that of [2]. Its finite- k inequality, evaluated exactly (dynamic programme over r , all truncation tails added), gives $\frac{1}{k} \log(\text{RHS}/k!) = -4.0 \times 10^{-4}$ at $k = 30001$ and -6.8×10^{-4} at $k = 100001$ for $\lambda = 1.004$ (it is positive at $k = 10^4$ because of the polynomial union-bound factor). The numerics are floating point, not interval arithmetic. Why the naive route fails and this one does not: the slot counts of different windows are increasing functions of a common random set, so FKG gives $\mathbb{E} \prod N_i \geq \prod \mathbb{E} N_i$ — the wrong direction — and negative association fails for nested value windows; but one never needs the product to factor, only a simultaneous *typical* upper bound (single-window Chernoff plus a union bound), and the price is the slack ε (the $\varepsilon = 0$ heuristic ceiling of the same computation is 1.0117).

2.5 Further refinements

Both parities. With the stable rule $I_{\text{even}} = \{i \text{ even} : b_i - 1 > k\}$, $I_{\text{odd}} = \{i \text{ odd} : b_i - 1 > k, i \pm 1 \notin I_{\text{even}}\}$, Lemma 1 still applies; the odd factors are 2-dependent, and a Hölder/Cauchy–Schwarz split (exponent $p = 1.6$) gives $\lambda_B = 1.000384$, rigorous except for the routine geometric-to-exponential

limit which we have not written out in full. The numerically exact rate of this two-parity encoding (transfer operator, confirmed by Monte Carlo to 5×10^{-7}) is $\lambda = 1.000437$; certifying it would require a verified top eigenvalue of a two-dimensional positive transfer operator. We regard ≈ 1.0005 as the ceiling of the “one value instead of one index” mechanism.

What did not help. Allowing runs of adjacent dropped indices never pays (the second index would need a window of more than $2k$ points); optimizing a threshold is pointless since the pointwise-optimal rule is exactly “include i iff $b_i - 1 > k$ ”.

3 Short superpatterns for $k = 7$ and $k = 8$

Theorem 2. $\text{sp}(7) \leq 23$ and $\text{sp}(8) \leq 30$. Witnesses: $\sigma_7 = 7\ 20\ 13\ 10\ 2\ 18\ 23\ 4\ 12\ 16\ 8\ 5\ 19\ 15\ 1\ 9\ 22\ 14\ 6\ 17\ 11\ 3\ 21$,

$\sigma_8 = 13\ 4\ 25\ 18\ 8\ 30\ 12\ 22\ 1\ 28\ 19\ 10\ 6\ 14\ 24\ 27\ 3\ 16\ 21\ 11\ 5\ 20\ 29\ 15\ 7\ 23\ 2\ 17\ 26\ 9$.

Two further 7-superpatterns of length 23: $7\ 18\ 12\ 2\ 15\ 23\ 6\ 19\ 13\ 4\ 8\ 11\ 22\ 17\ 3\ 20\ 10\ 5\ 14\ 1\ 21\ 9\ 16$ and $10\ 16\ 2\ 22\ 6\ 11\ 19\ 15\ 5\ 21\ 12\ 1\ 9\ 18\ 4\ 13\ 23\ 7\ 17\ 3\ 14\ 20\ 8$.

How found. Simulated annealing over S_n with moves = adjacent-position and adjacent-value swaps, energy = #missing patterns + $0.4 \cdot$ #patterns realized exactly once, with incremental re-evaluation (only subsets containing a swapped position change). Length 24 for $k = 7$ was found in under a second by essentially every chain; length 23 within a minute in 3 of 4 chains; length 22 was not found (12 chains, best 9 missing patterns after 2×10^5 moves each). For $k = 8$, length 32 was found after 675 moves length 31 after 3087 moves (about a minute), and length 30 with a second search variant (dynamic weights on persistently missing patterns plus occasional insertion moves); length 29 was not found (best 12 missing). An 8-superpattern of length 31 is $25\ 21\ 11\ 4\ 27\ 8\ 17\ 19\ 1\ 13\ 31\ 5\ 23\ 16\ 26\ 10\ 6\ 29\ 15\ 20\ 28\ 3\ 9\ 14\ 24\ 2\ 12\ 30\ 18\ 7\ 22$, and one of length 32 is $13\ 26\ 3\ 31\ 11\ 16\ 23\ 8\ 20\ 12\ 30\ 6\ 18\ 27\ 7\ 1\ 21\ 29\ 15\ 4\ 9\ 22\ 32\ 19\ 2\ 14\ 24\ 10\ 28\ 5\ 17\ 25$.

Verification. (i) Independent brute force in Python (enumerate all $\binom{23}{7} = 245,157$ resp. $\binom{30}{8} = 5,852,925$ subsets, collect patterns, compare count to $k!$): both witnesses contain all 5040 resp. 40320 patterns. (ii) Lean 4: see §6.

Structure. Every solution found has $\text{LIS} = \text{LDS} = k$ exactly (the forced minimum), short irregular runs (lengths 1–4), and no visible zigzag or grid structure; the solutions are not “word plus tie-breaking” objects. Indeed ζ_k is *irredundant*: for $k = 5, 6, 7$ deleting any single entry of the Engen–Vatter permutation destroys universality, so short superpatterns are not trimmed zigzags. The patterns that resist longest at $n = 22$ ($k = 7$) are near-monotone or layered (1234567 , 2134567 , 2176543 , 7123456 , ...), consistent with the analysis in [3] where layered permutations are the hard case.

Circular superpatterns. Manjunath and D’Souza [9] define circular k -superpatterns (patterns up to rotation) and prove $L_{\text{circ}}(k) \leq \text{sp}(k - 1) + 1$ together with a zigzag construction of length $\lceil ((k - 1)^2 + 1)/2 \rceil + 1$ for odd k . Theorem 2 therefore gives $L_{\text{circ}}(8) \leq 24$ and $L_{\text{circ}}(9) \leq 31$, improving their 25 and 34. Their closed form for Miller’s score function is equivalent to the run-change identity used in §4 to show that zigzag-type constructions cannot beat $k^2/2$.

Context. We now have $\text{sp}(6) = 17$, $19 \leq \text{sp}(7) \leq 23$ and $20 \leq \text{sp}(8) \leq 30$ (the lower bounds from $\text{sp}(k + 1) \geq \text{sp}(k) + 1$, obtained by deleting the maximum entry of a $(k + 1)$ -superpattern and using the pattern $\pi \oplus 1$), versus $\lceil (k^2 + 1)/2 \rceil = 19, 25, 33$; note $30 < 32 = 8^2/2$. Together with

$\text{sp}(6) = 17 < 18 = 6^2/2$, this is mild evidence against the conjecture $\text{sp}(k) \sim k^2/2$ of [10] (Hunter [5] announced, without proof, $\text{sp}(k) \leq \frac{15}{32}k^2 + O(k)$).

4 Structured constructions and what cannot beat $k^2/2$

Engen–Vatter’s construction is a word over $[k+1]$ (the zigzag) whose ties are broken so that equal letters form a decreasing subsequence. Their ζ_k is *irredundant*: for $k = 5, 6, 7$ no single entry can be deleted without losing a pattern. Nevertheless the tie-breaking idea is much more general than the zigzag:

Observation. Arnarson’s record 6-superpattern of length 17 is exactly the decreasing-tie-break of the word $3\ 6\ 5\ 1\ 6\ 7 \mid 3\ 4\ 2\ 6\ 5\ 7 \mid 1\ 4\ 6\ 3\ 7$ over $[7] = [k+1]$: three near-permutation blocks with non-uniform letter frequencies. (Its inverse splits into 7 decreasing chains of consecutive values.)

Searching this family (“ B distinct blocks over $[k+1]$, decreasing ties”) gives, all verified by two independent checkers:

- a 6-superpattern of length 18 of the form $\rho\rho$ with $\rho = 1\ 4\ 8\ 5\ 7\ 3\ 9\ 2\ 6 \in S_9$;
- a structured 7-superpattern of length $24 = (k^2 - 1)/2$: word $3\ 7\ 1\ 5\ 2\ 8\ 4\ 6 \mid 2\ 4\ 8\ 1\ 5\ 3\ 6\ 7 \mid 2\ 4\ 7\ 5\ 1\ 8\ 3\ 6$, permutation $9\ 21\ 3\ 15\ 6\ 24\ 12\ 18\ 5\ 11\ 23\ 2\ 14\ 8\ 17\ 20\ 4\ 10\ 19\ 13\ 1\ 22\ 7\ 16$. Three *identical* blocks ρ^3 , $\rho \in S_8$, never work (exhaustive: at most 5039/5040, always missing 7654321 — the same even-length phenomenon as in [3]).

For $k = 8$, three blocks over $[10]$ (length 30) reach 40316/40320, missing only 54367812, 78123456, 78123546, 87654132; three blocks over $[9]$ (length 27) fail badly. The unstructured witnesses σ_7, σ_8 of §3 are **not** in this family: their minimal “row alphabet” (over all eight symmetries) is 9–10 for $k = 7$ and 13 for $k = 8$, with mixed tie directions. So the family is a useful search space but the true optima leave it.

Negative results (elementary; proofs in the project log). (N1) Any “copy model”, in which each value of the pattern has a fixed candidate set of letters, collapses to a word over $[k]$ containing all permutations literally and hence has length $\geq k^2 - o(k^2)$ by Kleitman–Kwiatkowski. (N2) If the words $\sigma|_Y$ (Y a k -set of values) take only polynomially many distinct values, Hunter’s bound on $F(k, n)$ forces length $\geq (1/2 - o(1))k^2$; so any construction whose freedom is a threshold split or a bounded number of shifts cannot beat $1/2$. (N3) Every zigzag-type word (monotone runs over residue classes, any alphabet, any run directions, mod- q variants, with or without merged runs) needs $\geq k - 1$ run changes to embed the alternating permutation $1\ 3\ 2\ 5\ 4\ \dots$, hence length $\geq (1/2 - o(1))k^2$; merging two zigzags on low/high values reproduces the automaton analysed by Hunter [5] and gains only $O(k^{3/2})$. The common reason: gains must serve the alternating permutation ($k - 2$ extrema) and the near-monotone permutations simultaneously, and realizations by shifts and monotone-row merges cannot do both. One might hope that rows (value classes) whose internal tie-break pattern is non-monotone could help. For identical blocks this is provably not so: in the family ρ^B over an alphabet of size $m = k + d$ with arbitrary row patterns, a merged value costs a block exactly like a descent, so the identity and the decreasing permutation cost $k - \max_S \text{asc}(S)$ and $k - \max_S \text{desc}(S)$ blocks, both maxima are attained at $S = [m]$, and $\text{asc}(\rho) + \text{desc}(\rho) = m - 1$; hence $B \geq (k - d + 1)/2$ and the length is at least $(k - d + 1)(k + d)/2$, i.e. $(1/2 - o(1))k^2$ whenever $d = o(k)$. Consistently, the record witnesses embed every hard family (monotone, alternating, layered) using monotone fibres only; their few non-monotone rows are single-element displacements that mop up irregular patterns. Whether *non-identical* blocks, or alphabets $(1 + \delta)k$ with $\delta = \Theta(1)$, can beat $1/2$ remains open.

5 Exact values for words over $[k + 1]$

Let $f(k; r)$ be the least length of a word over $[r]$ containing every $\pi \in S_k$ as an order-isomorphic subsequence; $f(k; r) \geq \text{sp}(k)$, and $f(k; k + 1) \leq (k^2 + k)/2$ [4].

Theorem 3. $f(2; 3) = 3$, $f(3; 4) = 6$, $f(4; 5) = 10$, $f(5; 6) = 15$; i.e. $f(k; k + 1) = (k^2 + k)/2$ for $2 \leq k \leq 5$. Moreover $f(3; 5) = f(3; 6) = 5 = \text{sp}(3)$, $f(4; 6) = f(4; 7) = 9 = \text{sp}(4)$, and $f(5; 7) \geq 13 = \text{sp}(5)$.

Witnesses: 232; 241323; 2435142314; 135642135642135 (Miller’s word); 25314; 314625314; 371426315.

Method. SAT encoding (CaDiCaL via PySAT): one-hot letter variables $x_{i,\ell}$; for every target word $t = s \circ \pi$ ($\pi \in S_k$, $s : [k] \rightarrow [r]$ increasing) reachability variables $R_{i,j}$ (“ $t_{1..j}$ embeds in $w_{1..i}$ ”) with clauses $R_{i,j} \rightarrow R_{i-1,j} \vee (R_{i-1,j-1} \wedge x_{i,t_j})$, and one clause per π requiring $\bigvee_s R_{n,k}$. Symmetry breaking by complementation ($w_1 \leq \lceil r/2 \rceil$). Lower bounds are UNSAT results ($n = 13$: 19 s, $n = 14$: 166 s for $k = 5, r = 6$); every SAT witness was re-verified by an independent greedy embedding checker; the encoding reproduces Newey’s $f(3; 3) = 7$, $f(4; 4) = 12$.

Remarks. This answers Hunter’s Problem 3 affirmatively for $k \leq 5$. Note $f(k; k + 1) > \text{sp}(k)$ strictly for $k = 3, 4, 5$, while one more letter already recovers $\text{sp}(k)$ for $k = 3, 4$. The counting bound of [3] is slack at length $(k^2 + k)/2 - 1$, so the UNSAT proofs are genuinely structural. $f(6; 7) \in [17, 21]$ remains open (the $n = 17$ instance did not finish).

6 Gupta’s rosaries

A *rosary* for n is a cyclic word over $[n]$ such that every $\pi \in S_n$ occurs as a subsequence of some rotation read once around; $r(n)$ is the least length. Gupta conjectured $r(n) \leq n^2/2$ [11]; Lecouturier–Zmiaikou proved it for even n and showed $r(n) < n^2/2 + n/4 - 1$ for odd n [6].

Theorem 4. $r(2) = 2$, $r(3) = 4$, $r(4) = 8$, $r(5) = 12$ (witnesses 12, 1232, 12341232, 123432152435); $r(6) \in \{16, 17\}$; and for odd $n \in \{7, 9, 11\}$ the word

$$R_n = (1\,2\,\dots\,n)^{\lfloor m/2 \rfloor} (1\,2\,\dots\,m+1) (n\,\dots\,2\,1)^{\lceil m/2 \rceil - 1} (n\,\dots\,2), \quad m = \frac{n-1}{2},$$

of length exactly $\lfloor n^2/2 \rfloor$, is a rosary. Consequently Gupta’s conjecture holds for all $n \leq 11$.

Method. Exact values by SAT on the cyclic word (start position carries $\pi(1)$; symmetry breaking by relabeling and rotation); $r(6) \geq 16$ from UNSAT at 14, 15. R_n is the word of [6, Thm 2] with its partial block shortened; verified by an independent C checker (DFS over permutation prefixes carrying the live set of start positions; $n = 11$: 7 s).

Dead end, recorded deliberately. R_n is **not** a proof route: [6] state that this word fails at $n = 21$ and exhibit a permutation for $n = 33$; we re-verified that their permutation is not contained in R_{33} . The hard permutations are near-alternating with all decreasing-block minima above $(n + 1)/2$. A proof for odd n needs a different construction.

7 Random permutations (data)

Let $t(n)$ be the least m such that at least half of S_m are n -superpatterns [1]; [3] report $t(n) = 1, 3, 7, 13, 20, 28, 36, 48$ for $n \leq 8$. Our sampling (300 permutations per length) gives, for $n = 8$:

fraction of superpatterns 0.14 at $m = 44$, 0.59 at 48, 0.91 at 53, consistent with $t(8) = 48$. For $n = 9$ (60 samples per length, fixed seed) the fractions are 0.43, 0.53, 0.75, 0.80 at $m = 58, 60, 62, 64$, giving $t(9) \approx 60$, and 20 samples per length for $n = 10$ give $t(10) \approx 70$ –72 (new values; we also find $t(7) = 37$ rather than the sampled 36 of [3]: 0.41 ± 0.02 at $m = 36$ versus 0.55 at $m = 37$ with 1000 samples each; $t(9)/81 \approx 0.74$, in line with $t(n) \approx 0.75n^2$ over this range and far above $n^2/4$). The patterns that appear *last* near threshold are not monotone: for $n = 8$, $m = 44$ the most frequently missing patterns are 13264875, 13248567, 21435678, 13254768 — identities perturbed by a few adjacent transpositions. This is consistent with the analysis of He–Kwan’s proof [7], where the $\log \log k$ loss comes entirely from the union bound over “structured maps” (unions of few increasing runs), and suggests these hybrid patterns are the real obstacle to Alon’s conjecture $t(n) \sim n^2/4$. The same ordering of difficulty (indecomposable $<$ monotone $<$ extremal) appears, in a different model, in the online-embedding results of Altschuler, Dubroff and Tikhomirov [12].

A falsifiable test of Alon’s conjecture. For $\tau \in S_j$ let $L_\tau(N)$ be the largest L such that $\tau^{\oplus L}$ (the direct sum of L copies) is contained in N uniform random points, and $c_\tau = \lim L_\tau(N)/\sqrt{N}$ (which exists by superadditivity). Alon’s conjecture implies $c_\tau \geq 2/j$ for every τ : otherwise the pattern $\tau^{\oplus L}$ of length $k = jL$ would need more than $(1/4 + \varepsilon)k^2$ points. Monotone τ have $c_\tau = 2/j$ exactly (from $\text{LIS} \sim 2\sqrt{N}$), and in general $c_\tau \leq \min(2/\text{lis}(\tau), 2\text{lds}(\tau)/j, e/j)$ (Greene’s theorem with Baik–Deift–Johansson for the first two bounds, a first-moment count for the third). Using an exact dynamic programme for $L_\tau(N)$ (validated against brute force on 1824 point sets $\times$ 152 patterns with no mismatch), we estimated c_τ for all 64 symmetry classes of τ of length 2 to 5 at N up to 10^5 . The paired statistic $jL_\tau - \text{LIS}$ on the same point set has mean in $[-5.3, 2.3]$ at $N = 25000$ ($\text{LIS} \approx 300$) for every τ , i.e. $c_\tau = 2/j$ to within about 0.01, with the z -scores of the 58 best-sampled classes spread exactly as noise. **There is no counterexample candidate**, and the data support the sharper statement $c_\tau = 2/j$ for all τ — equivalently, that a “ τ -Hammersley” growth process has the flux relation of the Hammersley process at intensity N/j^2 . We could not prove this: the known proofs of the LIS constant rely on exact solvability. Finally, for the hybrid patterns of length $k = 8, \dots, 32$ (identity with spaced adjacent transpositions, and $(21)^{\oplus k/2}$) the half-probability containment threshold equals that of the identity within $\pm 3\%$, and $N_{\text{id}}/(k^2/4) = 1.57, 1.43, 1.38, 1.34, 1.30, 1.23$ for $k = 8, 12, 16, 20, 24, 32$, decreasing like the Tracy–Widom $N^{1/6}$ correction — consistent with the ratio tending to 1.

7.1 Rigorous bounds on c_{21} , and a universal absence threshold

The first open case of the conjecture $c_\tau = 2/|\tau|$ is $\tau = 21$: is the layered pattern $2143 \dots (2L)(2L-1)$ contained in $(1 + \varepsilon)L^2$ random points w.h.p.? Previously known: $0.40 \leq c_{21} \leq e/2$ (a diagonal greedy and the first moment). We prove $0.787 \leq c_{21} \leq 1.140$.

Lower bound. Tile the square with cells of side $\delta/\sqrt{N}$; cells containing a copy of $\tau^{\oplus t}$ are i.i.d. Bernoulli($p_{\tau,t}(\delta^2)$), and a strictly increasing chain of such cells is a τ -chain. The strictly increasing chain in i.i.d. Bernoulli(p) sites on an $n \times n$ grid has length $\sim n \cdot 2\sqrt{p}/(1 + \sqrt{p})$ (Seppäläinen 1997), so $c_\tau \geq \sup_{\delta,t} t g(p_{\tau,t}(\delta^2))/\delta$. For $\tau = 21$, $t = 1$ ($p = 1 - e^{-\lambda} I_0(2\sqrt{\lambda})$, $\lambda = \delta^2$) gives 0.577 at $\delta = 1.11$, and $t = 2$ (cells containing 2143, exact counts) gives 0.598. The same construction gives $c_\tau \geq 0.55$ –0.60 $\cdot (2/|\tau|)$ for all τ of length ≤ 5 , and $c_\tau \geq (0.7 - o(1))/(|\tau|\sqrt{\ln |\tau|})$ in general; it cannot exceed $0.6 \cdot 2/|\tau|$ by construction (one block per cell).

Better lower bound: a renewal sweep. Work with a unit-intensity Poisson process on the quadrant. From a corner c sweep the squares $c + (0, s)^2$ with s increasing and stop at the first arrival that completes a copy of 21 inside the swept square; the earlier points then form an increasing chain, the

new point b lies on the top or right edge, and we take as partner the chain point a with the least coordinate that forms a descent with b . The corner of the copy, $c' = \text{coordinatewise max of } \{a, b\}$, dominates b , and since b is on the boundary of the open swept square, the quadrant above-right of c' is disjoint from the swept region. Hence the increments $(dx, dy) = c' - c$ of successive steps are i.i.d., and in N points of the unit square the chain of copies has length $(2 + o(1))\sqrt{N}/\mathbb{E}[dx + dy]$, i.e. $c_{21} \geq 2/\mathbb{E}[dx + dy]$ (this is Hammersley's 1972 argument for the LIS, applied to descent pairs). Conditioned on the stopping radius r and on n earlier points, those points are i.i.d. uniform in $(0, r)^2$ conditioned to be a chain, so their sorted x - and y -coordinates are independent uniform order statistics, $P(n) = n/(n+1)!$, $\mathbb{E}r = \Gamma(n + \frac{3}{2})/n!$, and the cost $dx + dy = r + x_a$ has the exact series

$$\mathbb{E}[dx + dy] = \sum_{n \geq 1} \frac{\Gamma(n + \frac{3}{2})}{n!(n+1)!} \sum_{k=0}^{n-1} \left(1 + \frac{k+2}{n+2}\right) = 2.5425687 \dots,$$

giving $c_{21} \geq 0.7866$. (Series recomputed and the rule simulated independently: 2.5434 ± 0.002 from $4 \cdot 10^5$ steps. A strip-shaped sweep gives the closed form $\sqrt{2e(4-e)} = 2.6397$, i.e. 0.7577; sweeping $\ell_{1.5}$ -balls gives 0.803 in simulation but no closed form.) The same sweep works for any $\tau \in S_j$: conditioned on the stopping radius the earlier points are a uniform τ -avoiding permutation, so $\mathbb{E}[dx + dy]$ is an exact series in $|\text{Av}_m(\tau)|$ and first-occurrence counts (enumerated for $m \leq 11$, tails bounded via the Catalan, Gessel and Bóna formulas). This gives $c_\tau \geq 0.535\text{--}0.552$ for all $\tau \in S_3$ (cell construction: 0.389; conjecture 2/3) and $c_\tau \geq 0.385\text{--}0.413$ for all $\tau \in S_4$ (cell construction: 0.30; conjecture 1/2); the square-rule cost was re-simulated independently for 123, 231 and 21 and agrees with the series to 0.3%. The method is intrinsically capped: any renewal rule pays at least the cheapest descent pair in the quadrant, whose mean cost is 2.386 (simulation), so it cannot exceed $c_{21} \approx 0.84$; the remaining gap to the conjectured 1 is the same 0.80 ratio by which Hammersley's greedy ($\sqrt{8/\pi}$) misses the LIS constant 2, and closing it needs correlation between consecutive blocks rather than a better local rule.

Upper bound: canonical copies. Among all copies of $\pi \in S_k$ in Π_N take one minimizing $\sum_r x_r$; then for every r the strip $S_r = (x_{r-1}, x_r) \times (y_{(\pi(r)-1)}, y_{(\pi(r)+1)})$ contains no point (else moving p_r there gives a copy with smaller $\sum x$), and the strips are pairwise disjoint. By the Mecke formula,

$$\mathbb{E}\#\{\text{canonical copies}\} = N^k \int_{\Delta_x \times \Delta_y} \exp\left(-N \sum_r g_r(h_{\pi(r)-1} + h_{\pi(r)})\right) dg dh,$$

in x -gap and y -gap coordinates, and permuting the x -gaps shows that **this quantity is the same for every** $\pi \in S_k$ (it refines the identity $\mathbb{E}\#\text{copies} = \binom{N}{k}/k!$). A Chernoff extension of both simplices to orthants, integration of the x -gaps, and the scaling $h = u/\sqrt{N}$, $s = t = \sigma\sqrt{N}$ leave a one-dimensional positive transfer operator $(T_\sigma f)(u) = \int_0^\infty e^{-\sigma u'} f(u')/(\sigma + u + u') du'$ and $\mathbb{E} \leq N^{-1/2} e^{2\sigma\sqrt{N}} \int e^{-\sigma u_0} (T_\sigma^k 1)(u_0) du_0$. A Collatz–Wielandt test function ($\sigma = 1.5$, $\varphi(u) = 10^{-3} + (1 + u/1.7777)^{-0.9584}$, $\sup T\varphi/\varphi \leq 0.26808$, recomputed independently) gives, for **every** $\pi \in S_k$,

$$\Pr(\pi \subseteq \Pi_N) \leq 668 N^{-1/2} \exp(3\sqrt{N} - 1.3163 k).$$

Hence every pattern of length $k > 2.279\sqrt{N}$ is absent w.h.p. (the first moment gives $e\sqrt{N} = 2.718\sqrt{N}$), $c_\tau \leq 2.279/|\tau|$ for all τ , $c_{21} \leq 1.140$, and the containment threshold of *every individual* $\pi \in S_k$ is at least $(0.1925 - o(1))k^2 = (1.42/e^2)k^2$, versus k^2/e^2 from counting. For the identity the same bound says $\text{LIS} \leq 2.279\sqrt{N}$ (truth 2), so the method is about 14% off in both cases; canonical rules tailored to the 21 structure did worse than the universal one. (This does not give a deterministic superpattern bound: the number of canonical k -subsets of a fixed permutation is

not controlled.) The spectral-radius certificate is floating-point quadrature, not interval arithmetic; the rest is exact.

Is the identity the easiest pattern? No — it is the hardest. With a containment tester using two-dimensional bounds consistency and per-cell LIS/LDS pruning (validated against brute force on 56,000 samples and against an LIS oracle for the identity; `experiments/w21-threshold-numeric/`) we measured half-probability thresholds $n_{1/2}(\pi)$ with 600–4000 samples per point up to $k = 40$. In units of k^2 : identity (from the LIS, 10^5 samples) 0.362, 0.345, 0.332, 0.323, 0.316, 0.311, 0.307, 0.304 and random patterns (mean of 4–8 per k , spread 0.3–1%) 0.347, 0.323, 0.307, 0.295, 0.286, 0.278, 0.272, 0.268 for $k = 12, 16, \dots, 40$. The ratio random/identity decreases monotonically $0.959 \rightarrow 0.881$; the identity’s excess over $k^2/4$ grows like $0.6k^{4/3}$ (Tracy–Widom), whereas the random patterns’ excess has stopped growing (28.2, 29.0, 28.9, 27.9 at $k = 28, \dots, 40$); and the log–log slope of $n_{1/2}/k^2 - c$ against k is $-0.62 \approx -2/3$ for the identity at $c = 1/4$ (control) but -1.3 to -1.8 and steepening for random patterns at $c = 1/4$, becoming $-2/3$ only at $c \approx 0.20$ (bias-corrected ≈ 0.23). Every fitted form puts the random-pattern limit in $[0.19, 0.23]$. Structured patterns sort into two families: the layered pattern $(21)^{k/2}$ and the decreasing pattern are exactly as hard as the identity, while $\sqrt{k} \times \sqrt{k}$ tilted grids sit with the random patterns (350.4 vs 351.8 at $k = 36$). The one escape not excluded by finite k — a correction that is positive at $k \leq 40$ but eventually negative — would require the random thresholds to cross $k^2/4$ from above near $k \approx 60$; a run at $k = 48$ (feasible) would test it. Taken at face value: the constant $1/4$ in Alon’s conjecture is the *maximum* over patterns, attained by the monotone and layered ones; random patterns have limit $\approx 0.22 \pm 0.02$; and the universal absence constant satisfies $\kappa_{\text{univ}} \approx 2.13 > 2$, so $c_\tau \leq 2/|\tau|$ cannot come from a π -uniform bound. On the theory side, a second level of the lex-min hierarchy (two-point moves) lowers the identity’s constant to ≈ 2.24 (uncertified) and mixed leftmost/lowest rules are worse (2.35), so the universal constant 2.279 stands.

7.2 Towards Alon’s conjecture: a larger class at $n = O(k^2)$

He and Kwan [7] prove that a random permutation of length $2000k^2 \log \log k$ is a k -superpattern, and that length $20k^2$ suffices for the *quasirandom* patterns $Q_k = \{\pi : L_\Delta(\pi) \leq 3\sqrt{k} \text{ for all } \Delta\}$, where $L_\Delta(\pi)$ is the length of the longest Δ -shift chain (the LIS of the partial map $i \mapsto \pi^{-1}(\pi(i) + \Delta)$). Their argument reduces to a random $2k \times m$ Bernoulli($\frac{1}{2}$) matrix M ($m = n/4k$) and runs ℓ greedy “threads” at row shifts $t_1 < \dots < t_\ell$; two threads share exposed cells only along Δ -shift chains, and they bound the sharing by $L_\Delta \cdot (\text{maximal zero-run in } M) \leq L_\Delta \log^2 k$. We sharpen both halves of that bound.

Lemma 3. For shifts $t < t + \Delta$, with (z_a) the zero-run profile of the leader thread, the number of shared cells is at most $W_\Delta(z) = \max_A \sum_{a \in A} (z_a + 1)$ over Δ -shift chains A of elements reached by the leader.

Theorem 7. Let $\mathcal{A}_{L,s}$ be the event that every set of at most L cells in distinct rows has total zero-run less than s (so $\Pr(\neg \mathcal{A}_{L,s}) \leq 2(2km)^L e^{-s/8}$ for $s \geq 4L$). If the pairwise shifts satisfy $L_{t_j-t_i}(\pi) \leq L$, $s \geq 4L$ and $(\ell-1)(s+L) \leq m/2$, then

$$\Pr(M \not\supseteq \pi \text{ and } \mathcal{A}_{L,s}) \leq \Pr(\text{Bin}(\ell(m/2 - k), \tfrac{1}{2}) \leq \ell(k-1)).$$

The proof makes He–Kwan’s device of “pretending that $\mathcal{A}$ holds” precise: a cell is pretended to be a 1 (without exposure) exactly when exposing it as 0 would violate the run-sum condition; on $\mathcal{A}$

every pretended cell is a genuine 1, the modified threads coincide with the real ones, and Lemma 3 (which uses only the interval structure of the threads) bounds each pairwise overlap by $s + L$; the really exposed cells then number at least $\ell(m/2 - k)$ with at most $\ell(k - 1)$ ones, and the exposure principle applies.

Theorem 8. For large k a uniformly random permutation of length $72k^2$ contains w.h.p. every $\pi \in S_k$ for which fewer than $k/(3 \ln k)$ shifts Δ have $L_\Delta(\pi) > k/(3 \ln^2 k)$. The exceptional set has size at most $k! e^{-(1/3-o(1))k/\ln k}$.

Proof. Take $m = 18k$, $\ell = \lceil \ln k \rceil$, $L = \lfloor k/(3 \ln^2 k) \rfloor$, $s = \lceil 20L \ln k \rceil$; then $\Pr(\neg \mathcal{A}_{L,s}) \leq 2 \exp(L(2 \ln k + 3.6) - 2.5L \ln k) \rightarrow 0$. For such π choose ℓ shifts in $\{0, \dots, k\}$ greedily so that no pairwise difference lies in the set F of bad shifts (each choice forbids $2|F| + 1 < 2k/(3 \ln k) + 1$ values, so ℓ choices fit). Theorem 7 applies since $(\ell - 1)(s + L) \leq 7k + \ln k \leq m/2$, and gives $\Pr \leq \Pr(\text{Bin}(8\ell k, \frac{1}{2}) \leq \ell k) \leq e^{-8\ell k D(1/8 \| 1/2)} \leq k^{-2.5k}$; a union bound over k^ℓ patterns finishes, and He-Kwan's coupling (their Lemma 2.2) transfers the statement from M to permutations of length $2 \cdot 2k \cdot m = 72k^2$. The size of the exceptional set follows from $\Pr(\max_\Delta L_\Delta(\pi) \geq L') \leq k(e^2 k/L'^2)^{L'}$ for uniform π [7, Lemma 3.1]. ■

Since $k/(3 \ln^2 k) \gg 3\sqrt{k}$, this strictly enlarges the class handled at $n = O(k^2)$: the patterns *not* covered number $k!e^{-\Theta(k/\ln k)}$ instead of $k!e^{-\Theta(\sqrt{k})}$, and He-Kwan's own tightness family (unions of $\log^{10} k$ increasing runs on value intervals) now lies inside the covered class. The full proofs are in `experiments/w9-alon-threads/proof.md`; Lemma 3 was also checked by simulation (no violations in 400 instances; numerically W_Δ is 6–8 times below He-Kwan's bound).

Removing the $\log k$ from the global event. The cells that matter in Lemma 3 are the leader's *start cells* along a shift chain; their columns increase with the element, so listed by column their row sequence is $\pi|_A$, and if $\text{LDS}(\pi|_A) \leq \rho$ they form a union of at most ρ monotone (row- and column-increasing) chains — a ρ -staircase set. The number of ρ -staircase sets of size j in $[2k] \times [m]$ is at most $\binom{2k}{j} \binom{m}{j} \rho^{2j}$ (choose rows and columns; a permutation with $\text{LDS} \leq \rho$ is determined by labelling positions and values with their increasing-subsequence class), i.e. $e^{O(j \log(\rho k/j))}$ rather than $(2km)^j$. Replacing $\mathcal{A}_{L,s}$ in Theorem 7 by the event “every ρ -staircase set of $\leq L$ cells has run-sum $< s$ ” (the proof goes through verbatim) lets one take $s = O(L \log(\rho k/L))$.

Theorem 5. Let $\mathcal{L}_r$ be the unions of at most r increasing runs on value intervals and $\mathcal{D}_r = \{\pi : \text{LDS}(\pi) \leq r\}$. For large k , a uniformly random permutation of length $800k^2$ contains w.h.p. every $\pi \in \mathcal{L}_r$ with $r \leq k/(2 \ln(k+1))$, and every $\pi \in \mathcal{D}_r$ with $r \leq k$, provided that fewer than $k/(2\ell_r) - \frac{1}{2}$ shifts Δ have $L_\Delta(\pi) > k/\lambda_r$, where $\ell_r = \lceil (\ln r + 1)/29 \rceil$ (resp. $\lceil (2 \ln r + 1)/29 \rceil$) and $\lambda_r = (\ell_r - 1)(\ln r + 10) = O(\ln^2 r)$. In particular there is **no condition at all** for $r \leq e^{28}$; using the two row-disjoint shifts $t = 0$ and $t = k$ (a k -shift chain is empty) this extends to $r \leq e^{57}$ for $\mathcal{L}_r$ and $r \leq e^{28}$ for $\mathcal{D}_r$ with no condition.

The proof (in `experiments/w13-global-event/proof.md`) uses ℓ_r threads and the staircase event with $\rho = r$; the union bound over the $\leq (k+1)^r r^k$ patterns of $\mathcal{L}_r$ (resp. r^{2k} of $\mathcal{D}_r$) is paid by $e^{-29\ell_r k}$ per pattern. For random interleaving words the hypothesis holds with room ($\max_\Delta L_\Delta \leq 1.5k/\sqrt{r}$ empirically), so this covers the “run-quasirandom” part of the residual class of Theorem 8, and at $r = k$ it reproduces Theorem 8 with threshold $14.5k/\ln^2 k$ in place of $k/(3 \ln^2 k)$. Numerically, riding between threads for run-unions is concentrated on shifts aligned with the run height (joint failure ratio 2.4 at $\Delta = \text{height}$, ≈ 1.0 – 1.2 elsewhere).

Speed- N bounds for bounded numbers of runs. For the union-of-runs class one can bypass threads entirely. Write such a pattern as π_w for an interleaving word $w \in [r]^k$ (letter j marks the

positions of run j). Two elementary facts: $\pi_w \subseteq P$ iff P contains r vertically *separated* chains whose merge word is w ; and every w is a subsequence of the periodic word $(12 \dots r)^{\rho(w)}$ with $\rho(w) = 1 + \#\{i : w(i+1) \leq w(i)\} \leq k$, so every union of r runs is a pattern of the periodic shuffle. For the periodic shuffle itself, tile the square into $4r^2(r+1)m^2$ boxes ($k = rm$) and call a box good if it is nonempty; a “molecule” is a choice of one good box in each of r consecutive box-columns, one per strip, and a chain of m molecules in the product order yields a copy. A Mirsky-type argument (the molecule poset has at least $n_0 n_1^r - n_1^{r-1} B$ elements and width at most $n_1^r + r n_0 n_1^{r-1}$, B the number of bad boxes) shows a chain of length m exists whenever $B \leq 2m^2$, and a Chernoff bound gives:

Theorem 9. For $r \geq 2$ and $N \geq 8(r+1) \ln(2er^2(r+1)) k^2$, $\Pr((12 \dots r)^{k/r} \not\subseteq \Pi_N) \leq e^{-N/(4r^2(r+1))}$; consequently, for every fixed r , a uniformly random permutation of length $n \geq 16r^2(r+1) \ln(2er^2(r+1)) k^2$ contains every union of at most r increasing runs on value intervals of length k with probability $1 - e^{-\Omega(n/r^3)}$.

The failure probability has speed N (like the LIS lower tail), which is what a union bound over the $e^{\Theta(k \log r)}$ residual patterns needs; the limitation is the threshold, linear in $r \log r$, so on the residual class $r \leq \ln^4 k$ this gives only $n = k^2 \text{polylog } k$. A matching lower bound in the fixed-strip model ($\Pr \geq e^{-N/r}$) shows that the speed is $N/\text{poly}(r)$ genuinely. Proofs in `experiments/w11-strips/proof.md`. *Certificate counting* (`experiments/w23-certificates/proof.md`) sharpens this: with a $3m \times 3rm$ grid of boxes per strip, $\Pr((12 \dots r)^{k/r} \not\subseteq_{\text{fix}} \Pi_N) \leq \exp(-\frac{N}{3r^2}(1 - 9r \ln(3er)/C))$ for $C > 9r \ln(3er)$, and the same for every block-grid pattern with threshold $9r \ln(3er^2)$ — a factor r better in speed than Theorem 9, but the $r \ln r$ threshold is intrinsic to the method: any “bad-box density $\leq f$ implies containment” lemma has $f \leq 1/r + o(1)$ (kill the boxes of strip 1 on the left half and of strip 2 on the right half), so the binomial tail is non-vacuous only when boxes hold $> \ln r$ points, i.e. $C > \ln r - o(1)$ for every Mirsky-type grid; and in the fixed-strip model the rate is at most $1/r$ for all C . For the identity the analogous Dilworth-cover certificate gives $\Pr(\text{LIS}(\Pi_N) < k) \leq (N+k)^k (e(k-1)^2/(N-k+1))^N$, rate $\ln(C/e)$, whose loss against the true Deuschel–Zeitouni rate is 0.60 at $C = e$ and $O(C^{-1} \ln C)$ for large C ; canonical (patience-sorting) covers cannot lower the e . Thus on Theorem 8’s residual class $r \leq \ln^4 k$ certificates give $n = O(k^2 r \ln r)$ with better constants but the same order, and reaching $O(k^2)$ there needs a certificate that encodes the r -dimensional Pareto front rather than a deterministic box tolerance.

Theorem 13 (speed N for a non-monotone pattern). Let $\pi = (12)^h = 1(h+1)2(h+2) \dots$, $k = 2h$, and let the N points carry i.i.d. fair colours (the two rigid value strips). Then

$$\Pr((12)^h \not\subseteq \Pi_N) \leq 2^{-N} \sum_{m=0}^N \binom{N}{m} \sum_{\substack{n=m \\ 2(k-1) \text{ parts}}} \frac{m!}{\prod_j (n_j!)^2} = \exp(-\rho(C)N + O(k \ln N)),$$

$$\rho(C) = \ln 2 - \sup_{0 < \beta \leq 1} \left[H(\beta) + \beta \ln \frac{4e}{\beta C} \right],$$

with $\rho(C) > 0$ iff $C > C_0 = 27.63$ ($\rho(50) = 0.17$, $\rho(100) = 0.31$, $\rho \uparrow \ln 2$, the exact cap of the two-colour model).

Proof idea (full proof: experiments/w26-pareto-ld/proof.md §1). Relax to *alternating chains*: a dominance chain whose colours alternate is a copy of $(12)^h$. Let $\lambda(p)$ be the longest alternating chain ending at p ; λ is monotone within each colour, and if no alternating chain of length k exists then $\lambda \leq k-1$. The certificate consists of the *generators* — the minimal elements of each level class $\{\lambda = t, \text{ colour } c\}$, which are antichains (probability $1/n!$ each) — with their labels; the nested

up-sets they generate cut the square, for each colour, into k slabs, and every non-generator of colour c and level t must lie in the intersection of the t -th slab of the other colour with the $(t+1)$ -st slab of its own. Those allowed regions have total area (summed over both colours) at most 1, because the two slab families are partitions of the square and the index pairs $(t, t+1)$ and $(t+1, t)$ are disjoint; so each non-generator is allowed with probability $\leq 1/2$. Summing over the choice and labelling of the generators gives the display; for the monotone word $0^h 1^h$ all classes are antichains and the certificate collapses to the Dilworth bound. ■

This is the first speed- N large-deviation bound for a pattern that is not a union of a bounded number of monotone runs handled by box tolerances, and it is k -uniform: it gives Conjecture LD (uniform speed N above threshold) for this pattern with an absolute, if large, constant. The relaxation is what costs the constant: the alternating-chain threshold is $\approx 9/16$ (alternating LIS $\approx 1.33\sqrt{N}$) against $\approx 1/4$ for the pattern, and dropping the witness and stacking constraints moves $9/16$ to 27.6 . Grid coarse-grainings of the exact three-dimensional Pareto sets cost $e^{\Theta(k^3)}$ and are useless; the same theorem for r strips gives $\text{cap } \ln(r/(r-1)) \approx 1/r$ (matching the fixed-strip cap) and thresholds $C_0(r) \approx 3.5r^3$ — optimal speed but a worse threshold than the box method, and nothing yet for the union bound over S_k . Exact numerics (tg.c) confirm speed N for $r = 2$: the rate $-\ln P/N$ at $C = 0.5$ is 0.027 at $k = 12$ and 0.026 at $k = 32$.

Theorem 14 (coarse-graining and Mirsky: speed N for chains of bounded gadgets). Cut $[0, 1]^2$ into a $g \times g$ array of cells, $N = mg^2$, and call a cell *good* if the standardised pattern of its points contains every $\tau \in S_b$; let $\bar{q} = \bar{q}_b(m) = \mathbb{E} q_b(n_c)$ with $n_c \sim \text{Bin}(N, g^{-2})$ (and $\bar{q} \geq 1 - e^{-m}$ for $b = 1$). (a) If $\pi = \tau_1 \oplus \dots \oplus \tau_h$ (or $\ominus$ throughout) with all $|\tau_i| \leq b$, then for $g = \Lambda h$ and $\Lambda > 2/\bar{q}$,

$$\Pr(\pi \notin \Pi_N) \leq \exp(-g^2 D(2/\Lambda \parallel \bar{q})) = e^{-\iota N}, \quad \iota = \frac{D(2/\Lambda \parallel \bar{q})}{m}, \quad N \geq m\Lambda^2 k^2,$$

where $D(a \parallel q) = a \ln \frac{a}{q} + (1-a) \ln \frac{1-a}{1-q}$. Instances: the identity ($b = 1$, $m = 3$, $\Lambda = 4$: $\iota = 0.277$ for $N \geq 48k^2$) and every layered pattern with layers of size ≤ 2 , in particular $(21)^{k/2}$ ($m = 6$, $\Lambda = 4$, $\bar{q} = 0.938$: $\iota = 0.122$ for $N \geq 24k^2$). (b) (Rigid rows.) For the block-grid class $\mathcal{G}(r, h)$ of Theorem 10, cut into $rn \times rn$ cells and call $(i, y) \in [n]^2$ good if all r^2 cells of column block i in row y of every value strip are nonempty; then with $m = 2 \ln r + 2$ points per cell and $n = 4h$,

$$\Pr(\pi \notin \Pi_N) \leq \exp\left(-\frac{N}{2.74 r^2 (2 \ln r + 2)}\right) \quad \text{for all } \pi \in \mathcal{G}(r, h), \quad N \geq 16(2 \ln r + 2)k^2,$$

and $m = \ln r + 2$ suffices for the tilted grid $(12 \dots r)^h$ itself.

Proof (full text: [experiments/w27-comparison/proof.md §2](#)). A chain of good cells strictly increasing in both coordinates yields a copy of π (cell i supplies τ_i ; distinct cells of the chain are in the $\oplus$ relation). If no such chain has length h , Mirsky's theorem covers the set E of good cells by $h-1$ antichains of $[g]^2$ under the strict product order, whose width is $2g-1$ (the $2g-1$ diagonals are chains covering $[g]^2$; $\{(1, y)\} \cup \{(x, 1)\}$ is an antichain), so $|E| \leq (h-1)(2g-1) < 2g^2/\Lambda$. The goodness indicators are nondecreasing functions of disjoint groups of the multinomial counts (with independent auxiliary coins), hence negatively associated (Joag-Dev–Proschan), and the Chernoff lower tail for $\text{Bin}(g^2, \bar{q})$ applies. For (b) the same chain, now in $[n]^2$, supplies one point per (block, strip) cell; the column offset $\tau_j(s)$ inside a column block realises the visiting order of block j , so one chain serves every $\pi \in \mathcal{G}(r, h)$ at once; $\Pr(\text{good}) \geq 1 - r^2 e^{-m}$ by a union bound over the r^2 cells, and $D(\frac{1}{2} \parallel 1 - e^{-2}) \geq 0.365$. □

Discussion. Part (a) is the first speed- N tail bound for $(21)^{k/2}$ and for layered patterns with bounded layers (Theorem 11's reserve greedy has failure speed $\leq \min(k, \text{strip height}) = 2$ for these,

so it cannot give it), and it uses no LIS structure: the cost is paid per cell, $\Theta(N)$ cells, each good with constant probability — a genuine *local* mechanism. The identity instance is a mechanism-free proof of the LIS lower tail with speed N (the Deuschel–Zeitouni rate is of course far better). Part (b) improves the periodic-word bound of §7 (threshold $9r \ln(3er)k^2$, speed $N/3r^2$) to threshold $16(2 \ln r + 2)k^2$ and extends it from the word to all of $\mathcal{G}(r, h)$, but the rate is still $\propto 1/(r^2 \ln r)$: one chain element costs r^2 cells of which only r are used, and for $r = h = \sqrt{k}$ the exponent at threshold is $\approx 5.8k$ — no better than Theorem 10 plus block splitting. This is the barrier in its cleanest form: the Mirsky mechanism needs the pattern to be a *chain* of $O(1)$ -cost gadgets (bounded blocks; bounded r), while a grid with $\min(r, h) \rightarrow \infty$ is a *lattice* of gadgets, whose constraint graph has no poset whose chains are the copies (whole strips are antichains, and antichain covers of the gadget poset number $e^{\Theta(N \ln k)}$). Two further exact facts (same file, §0–1): (i) *block splitting* $\Pr(\pi \not\subseteq \Pi_N) \leq \Pr(\pi \not\subseteq \Pi_M)^{\lfloor N/M \rfloor}$ shows that a bound $e^{-\iota k^2}$ at a single $N = C_0 k^2$, uniform in π , already gives Conjecture LD and Alon’s conjecture at $n = C_0 k^2$; the whole problem is therefore “exponent k^2 versus k at $N \asymp k^2$, uniformly in π ”. (ii) The naive comparison “identity hardest at every N ” is false already at $k = 4$: $\text{Av}_7(1324) = 2762 > 2761 = \text{Av}_7(1234)$ (Bóna; recomputed), the extremal pattern for $k \leq 8$ being $1 \oplus \text{dec}_{k-2} \oplus 1$ — exactly, $\text{Av}_9(14325) = 261863 > 261808$ and $\text{Av}_{10}(154326) = 3291662 > 3291590$ (our enumeration; note that a sequential-Monte-Carlo estimate of these deep tails is biased upward by up to a nat at $k = 6$, as 123654, Wilf-equivalent to the identity by Backelin–West–Xin, reveals) — and for $K > 1$ the comparison $\Pr(\pi \not\subseteq \Pi_N) \leq e^{ak} \Pr(\text{id} \not\subseteq \Pi_{N/K})$ is unfalsifiable at fixed k (Stanley–Wilf growth $L(\pi)^N$ against $(N/K)!/N!$), so the only content of the comparison principle is uniformity in k in the window $N \asymp k^2$, i.e. a comparison of per-point rates $I_\pi(C) = -N^{-1} \ln \Pr(\pi \not\subseteq \Pi_{Ck^2})$. The relabelling identity $\Pr(\pi \not\subseteq \Pi_N) = \Pr(\text{no increasing } k\text{-subsequence of } \Pi_N \text{ with value set } V \text{ such that } \rho|_V \cong \pi)$, valid for every $\rho \in S_N$, makes every π a *constrained* LIS problem with a constraint of density $1/k!$ among value sets — the union-bound entropy $k \ln k$ seen from the other side.

How slack is the union bound? (numerics.) For σ_n uniform let $M = \#\{\pi \in S_k : \pi \not\subseteq \sigma_n\}$; then $R(n, k) = \mathbb{E}[M \mid M > 0] = \mathbb{E}M / \Pr(M > 0)$ is exactly the factor by which the union bound overshoots. Exhaustive missing-pattern counts for $k = 6$ –9 (`experiments/w24-union-slack/`, 2000–30000 samples per point, the missing sets cross-checked against a naive enumeration; $k = 6$, $n = 28$ re-sampled independently here: $\Pr(M > 0) = 0.414$, $\mathbb{E}M = 2.47$, $R = 5.97$ against 0.414, 2.41, 5.81) give, at the median threshold $t(k)$ (where $\Pr(M > 0) = \frac{1}{2}$): $\ln R = 1.34, 1.93, 2.59, 3.26, 3.87$ for $k = 5, \dots, 9$ — successive increments 0.60–0.67, i.e. $\ln R \approx 0.65k \approx 0.2 \ln k!$; at $n = k^2$, $\ln R \approx 0.4$ – 0.5 ($k = 6, 7$), and $R \rightarrow 1$ by $n \approx 1.1k^2$. Given $M > 0$ the missing set is heavy-tailed ($\mathbb{E}M^2/(\mathbb{E}M)^2$ from 2 at $0.6k^2$ to $\geq 10^2$ near k^2) and clustered: pairs $\pi, \pi \circ (i \ i+1)$ co-miss 160–230 times more often than independence predicts at $t(7)$, dihedral images 20–25 times, random pairs 12–16 times, while reverse/complement pairs of the identity never co-miss (LIS and LDS are negatively related at fixed n); the missing set at $n \approx t(k)$ has 2–4 components under adjacent transpositions, the largest holding $\approx 70\%$. So at the true threshold the first-moment method is off by $e^{\Theta(k)}$ (not $e^{\Theta(k \ln k)}$): the union bound $\sum_\pi \Pr(\pi \not\subseteq \sigma_n) < 1$ would place the threshold where $\ln \mathbb{E}M = 0$, only $O(1) \cdot k$ nats — a factor $1 + O(1/k)$ in n/k^2 — above $t(k)$, but the slack is growing with k and a second-moment or clustering (hard-core) correction is needed to make it a proof.

Proposition 15 (the witness reduction; what a proof of the constant needs). With M , $\mu = \mathbb{E}M$, $R = \mu / \Pr(M > 0)$ as above and $\Lambda_\pi(n) = \mathbb{E}[M \mid \pi \not\subseteq \sigma_n]$ (the mean number of missing k -patterns of a *uniform* π -avoiding permutation of length n): (a) $R = \mathbb{E}[M \mid M > 0] \geq \max(1, \mu)$; (b) $R \leq \mathbb{E}M^2 / \mu = \mu^{-1} \sum_\pi p_\pi \Lambda_\pi \leq \max_\pi \Lambda_\pi$; (c) for any witness variable W on

$\{M > 0\}$, $R \leq \max_w \mathbb{E}[M \mid W = w]$. Hence if $\max_\pi \Lambda_\pi(n) \leq e^{ak}$ then $\mu e^{-ak} \leq \Pr(M > 0) \leq \mu$, and — provided $\ln \mu(n)$ decreases at rate $\Omega(1)$ per unit of n near the threshold (numerically 0.25–0.5; it is what speed- N per-pattern tails assert) — the median threshold $t(k)$ differs from the first-moment point $\{\mu = \frac{1}{2}\}$ by $O(ak) = o(k^2)$: **for $a = o(k)$ the threshold constant is the first-moment constant**, i.e. it is determined by the per-pattern rates $I_\pi(C)$ of Theorem 14’s discussion alone. *Proof.* (a) is the definition; (b) is Cauchy–Schwarz $\mu^2 \leq \mathbb{E}M^2 \Pr(M > 0)$ and $\mathbb{E}M^2 = \sum_\pi \mathbb{E}[M; \pi \not\subseteq \sigma_n]$; (c) is $\mu = \sum_w \Pr(W = w) \mathbb{E}[M \mid W = w]$. $\square$ This escapes the $\Delta \geq \mu^2/C$ cap of Janson-type bounds (§7, “What the standard concentration tools give”) because it bounds $\Pr(M > 0)$ from *below* through conditional expectations of a count, using no correlation inequality; indeed at $k = 6, 7$ Janson’s exponent $-\mu + \Delta/2$ is positive at every n with $\mu > 10^{-3}$ ($\Delta/\mu = 20$ –430) and the Bonferroni truncations of orders 2, 3 are negative resp. > 1 for all $n \leq 0.9k^2$, while (b) gives $\ln(\mathbb{E}M^2/\mu) = 3.0, 4.4, 5.5, 6.1$ against $\ln R = 1.8, 2.5, 3.1, 3.7$ ($k = 6, \dots, 9$ at $t(k)$; our sampler: 3.1 at $k = 6$). What the data say about *why* $R = e^{O(k)}$ (*experiments/w28-hardcore/*): there is no hard core — p_π varies over S_k by a factor ≤ 4 at $t(k)$, the top 1% of patterns carries 2% of μ , the hardest patterns are layered/co-layered (1543276, 7234561) rather than the identity, and $\text{corr}(p_\pi, \Lambda_\pi) = -0.75$ ($k = 6$), -0.43 ($k = 7$): hard patterns fail alone, generic ones in bulk. Given $M > 0$ the missing set is one diffuse component under one-point moves (92–94% of M in the largest component, vs 51–69% for a random set of the same size) but *not* the up-set of a missing $(k-1)$ -pattern (such up-sets explain $< 1\%$ of M for $M < 100$); R is made in the far tail (the share of $\sum M$ from events with $M \geq 256$ is 0, 0.07, 0.29, 0.48 for $k = 6, \dots, 9$), and the size of M is governed by *local emptiness* of σ_n , not by its LIS: Spearman correlation of M with the largest empty axis-parallel rectangle is +0.42 and with an empty 4×4 cell -0.29 , against +0.11 with LIS (6000 fresh samples, $k = 7$, $n = 37$; 98% of the σ with $M \geq 100$ have an empty 4×4 cell, 41% of those with $M = 0$). Two exact side facts: $\{\text{id} \not\subseteq \sigma_n\} \cap \{\text{rev} \not\subseteq \sigma_n\} = \emptyset$ for $n \geq (k-1)^2 + 1$ (Erdős–Szekeres; this is the observed zero co-miss count at $t(7) = 37$), worth only a factor 2; and below the threshold $R \geq \mu = e^{\Theta(k \ln k)}$, so the right conjecture is $R \leq \max(1, \mu) e^{O(k)}$, with content only in the window $\mu \in [e^{-O(k)}, e^{O(k)}]$. The combinatorial step a proof needs is thus a large-deviation bound on the *number* of patterns killed by one empty rectangle of area $\asymp 1/k^2$: that it is $e^{O(k)}$, not $e^{\Theta(k \ln k)}$. *This step, examined (experiments/w30-empty-rectangle/), turns out not to be defect-specific.* Exactly: $\pi \subseteq \sigma + x$ for a new point x iff $\pi \subseteq \sigma$ or x lies in a completion cell of a copy of some child $\pi^{(i)}$; so the revival region of a missing π is a union of empty rectangles, and a *uniformly random* new point revives exactly a fraction $1 - \mu(n+1)/\mu(n)$ of the missing patterns in expectation (≈ 0.41 at $k = 6$; measured 0.40), the centre of the largest empty rectangle revives 0.74–0.79 *independently of its area and aspect ratio*, and the best of nine grid points 0.96. There is no separate “kill count”: $K(Q) \approx 0.75M$ for every large empty Q , and the $e^{\Theta(k)}$ question is the tail of M itself. The one rigorous handle is strip deletion — conditionally on a $w \times h$ rectangle being empty, the standardisation of σ_n off its value (position) strip is uniform in S_{n-h} (S_{n-w}), so $\mathbb{E}[M \mid Q \text{ empty}] \leq \mu(n - \min(w, h))$ — together with $p_\pi(n-1) \leq n p_\pi(n)$ (prepend or append a new maximum), which for the typical largest empty rectangle ($\min(w, h) \asymp k\sqrt{\ln k}$) gives only $e^{O(k \ln^{3/2} k)} \mu$, or $e^{O(k\sqrt{\ln k})} \mu$ if the slope $-\partial_n \ln \mu$ is $O(1)$ uniformly in k (numerically 0.4–0.55; open). The statement must be about typical σ : adding one point to the length-17 6-superpattern revives $635 = 0.88 \cdot 6!$ of its 1538 missing 7-patterns.

The one-point slope (experiments/w33-slope/). Three exact identities govern $s_\pi(n) = \ln p_\pi(n-1) - \ln p_\pi(n) \geq 0$: $p_\pi(n)/p_\pi(n-1)$ equals the mean fraction of safe insertion slots of a uniform π -avoider of length $n-1$, equals $\mathbb{E}[S_{\max}]/n$ with $S_{\max}$ the number of positions accepting a new maximum (generating tree), and $p_\pi(n-1) - p_\pi(n) = \frac{1}{n} \mathbb{E}[|J_\pi(\sigma_n)|; \pi \subseteq \sigma_n]$ with J_π the points common to all copies. Since $S_{\max} \geq k-1$, $s_\pi(n) \leq \ln(n/(k-1))$ (improving $\ln n$ above), but

pointwise $O(1)$ remains open; what *is* proved is that a uniform lower bound $s_\pi \geq s_0$ over a window of length $\Theta(k^2)$ would already give speed- k^2 tails for every pattern (so it is as hard as the conjecture itself and, for μ , false: $\ln \mu \leq \ln k!$ is spent over the last $O(k \ln k)$ points), and that the window *average* of the μ -slope is $\leq H_0(1/\sqrt{C}) + o(1)$ (Deuschel–Zeitouni), i.e. $O(1)$. For the identity the slope has the closed form $s_{\text{id}}(Ck^2) \rightarrow \ln \frac{(4C+1)^2}{16C}$ (0.118, 0.288, 0.446 at $C = \frac{1}{2}, \frac{3}{4}, 1$), obtained from the Deuschel–Zeitouni rate $H_0(x) = -1 + \frac{x^2}{4} + \frac{x^2}{2} \ln \frac{4+x^2}{2x^2} + 2 \ln \frac{4+x^2}{4x}$ via $s = H_0 - \frac{x}{2} H'_0$ (we checked the algebra and the cubic zero $H_0(2-\varepsilon) \sim \varepsilon^3/12$); it vanishes at $C = \frac{1}{4}$, so at the point where a Tracy–Widom-type pattern reaches $p_\pi = 1/k!$ the slope is only $\Theta((\ln k/k)^{2/3})$. This sharpens the proviso in Proposition 15: the window there has length $\Theta(ak^{5/3}(\ln k)^{-2/3})$ rather than $O(ak)$, which is $o(k^2)$ iff $a = o(k^{1/3} \ln^{2/3} k)$ — still ample for the observed $\ln R \approx 0.7k$, i.e. $a \approx 0.7$. Numerically (SMC validated against exact enumeration for $n \leq 11$) every pattern of S_4 – S_7 tested has slope within $\pm 15\%$ of the identity’s at the same (k, C) , increasing in n , and the μ -slope at the threshold is 0.45–0.55 for $k = 6$ –9.

Theorem 16 (a typical pattern below $0.53k^2$: the gap-reserve greedy). Cut the values into $m = k/h$ strips of h consecutive values; a strip may carry an *arbitrary* sub-pattern $\sigma \in S_h$. Embed positions in order; when the next value v of a strip is to be placed, let $y'_L < y'_R$ be the heights of the placed strip points of nearest value below and above $(0, h$ if absent), m_b, m_a the numbers of unplaced values between them and v , and take the leftmost point (in x , beyond the global x -level) inside a window $W_t \subseteq [y'_L + \beta m_b, y'_R - \beta m_a]$ of height $\geq \beta$. (a) For every $\pi \in S_k$, $\Pr(\pi \notin \Pi_{Ck^2}) \leq \Pr(T_1 + \dots + T_m \geq k)$ with independent strip costs T_j , $\mathbb{E}T_j = C^{-1} \mathbb{E} \sum_{t \leq h} |W_t|^{-1}$, whose window law does not depend on C . (b) If π is uniform in S_k , the strip sub-patterns are i.i.d. uniform, so with $\Omega_h = \mathbb{E}[\sum_t |W_t|^{-1}]/h$ and $C > \Omega_h$: $\Pr(\pi \notin \Pi_{Ck^2}) \leq e^{-\eta k}$ over both π and Π , hence for all but an $e^{-\eta k/2}$ fraction of $\pi \in S_k$ individually, and a uniform σ_n with $n \geq (1+\varepsilon)Ck^2$ contains a uniform π with probability $1 - e^{-\Omega(k)}$. With the window centred at the proportional position of v in its gap, width 3, $\beta = 0.7$: $\Omega_{64} = 0.527$, $\Omega_{1024} = 0.515$ (Monte Carlo, C-independent; recomputed independently here to four digits), so a **typical pattern has threshold $\leq 0.527k^2$** against $0.757k^2$ for every pattern (Theorem 11). (c) (Barrier.) For every *value-blind* gap-window rule (windows measurable in the relative ranks seen so far), $\mathbb{E}T \geq (h+1)/(2C)$, so $\Omega_h \geq \frac{1}{2} + \frac{1}{2h}$: this family cannot reach $\frac{1}{2}$, let alone $\frac{1}{4}$.

Proof (full text: [experiments/w29-random-threshold/proof.md](#)). (a) The reserve β per unplaced value keeps every window of height $\geq \beta$ and every point inside its strip and strictly between its value-neighbours, so on $\{\sum X_p < k\}$ the chosen points form a copy of π ; leftmost searches in disjoint explored sets make the increments $X_t \sim \text{Exp}(C|W_t|)$, Y_t uniform on W_t given the past (the fresh-search lemma of Theorem 11), and the product form of the joint density gives independence across strips. (b) The standardisations of π^{-1} on the value blocks are i.i.d. uniform; Chernoff for the i.i.d. T_j ($\mathbb{E}e^{\theta T} < \infty$ for $\theta < C\beta$), Markov over π , and the Poisson-to-uniform coupling. (c) For a value-blind rule the rank r_t of the t -th value among the first t is uniform on $[t]$ and independent of the past; the $t-1$ placed points cut $[0, h]$ into t gaps $G_{t,j}$ with $\sum_j G_{t,j} = h$, the window lies in the r_t -th, and $\mathbb{E}[X_t | \text{past}] \geq (Ct)^{-1} \sum_j G_{t,j}^{-1} \geq t/(Ch)$ (harmonic–arithmetic); sum over t . $\square$

Discussion. This is the first rigorous separation of “typical” from “every” pattern: the reserve greedy of Theorem 11 needs strips that are monotone runs of π^{-1} , which for a random π forces strips of 2–3 values; here the strip may be anything, at the price that a random sub-pattern visits a uniformly random gap, of typical height h/t at time t , so the t -th point costs $\geq t/h$ whatever the window — the exact analogue of the fresh-quadrant barrier $\pi/8$ of Theorem 12(c), and the reason (c) stops at $\frac{1}{2}$. The rule actually used is *value-aware* (it centres the window at the proportional

position of v , which reveals future ranks), so (c) does not formally cover it; numerically it sits 0.015–0.03 above $\frac{1}{2} + \frac{1}{2h}$ for $h \geq 64$, and at $h = 2$ the value-aware optimum is $\approx 1.57 \geq 1.5$. The identity is *not* subject to the barrier (its gap is always the top one) and is handled by the corner rules; the two rules are complementary, and the same bound $0.53k^2$ holds for any fixed π whose strip sub-patterns have random-like cost. Reaching a constant below $\frac{1}{2}$ for random π — where the numerics say the truth is $\approx 0.22 < \frac{1}{4}$ — needs lookahead in x or a genuinely two-dimensional argument. End-to-end validation on real Poisson points at $k = 400$, $h = 40$ ($\Omega_{40} = 0.537$): success 2/20, 13/20, 20/20, 20/20 at $C = 0.5, 0.55, 0.6, 0.7$, each success asserting a copy of π .

Theorem 17 (below $\frac{1}{2}$: value-aware lookahead inside the gap). Keep the value strips of Theorem 16 but choose, among the points of the fresh half-strip $\{x' > a_s\} \times W$ above the gap of the value to be placed, the minimiser of $(x' - a_s) + \Phi(y')/C$, where $\Phi(y') = \bar{V}_{m_b}/(y' - y'_L) + \bar{V}_{m_a}/(y'_R - y')$ is a cost-to-go estimate for the m_b, m_a values still to be placed below and above, W is the gap shrunk by a relative margin ε on each side that still holds values, and $a_s = \max(a, e_j)$ is a *safe clock* (e_j = right edge of everything the strip has explored so far). (a) For uniform $\pi \in S_k$ and any nonnegative potentials $\bar{V}$: $\Pr(\pi \notin \Pi_{Ck^2}) \leq e^{-\eta k} + O_h(\ln k/k)$ whenever $C > \mathcal{V}_h/h^2$, where $\mathcal{V}_h$ is the rule's expected strip cost at intensity 1 in a unit-height gap. (b) $\mathcal{V}_n$ obeys the self-similar recursion $\mathcal{V}_n = \frac{1}{n} \sum_{m=0}^{n-1} \mathbb{E}[u'' + \mathcal{V}_m/y + \mathcal{V}_{n-1-m}/(1-y)]$, (u'', y) the minimiser of $x + \bar{V}_m/y + \bar{V}_{n-1-m}/(1-y)$ over a unit Poisson field on $(0, \infty) \times W$, with $\Pr(\min > s) = e^{-\text{area}(s)}$ in closed form; choosing $\bar{V}_n$ as certified upper bounds of the recursion (monotone rectangle rule) gives $\bar{V}_h/h^2 = 0.4765, 0.4680, 0.4649$ for $h = 64, 160, 512$ ($\varepsilon = 0.02$). Hence **a uniformly random k -pattern is contained in a random permutation of length $0.465k^2$ w.h.p.** (c) In the fresh-window model the rule with $\bar{V} = \mathcal{V}$ and $\varepsilon = 0$ is Bellman-optimal among all rules that commit each point inside the gap of its value using the fresh half-strip of that gap, with value $V_h/h^2 = 0.75, 0.663, 0.507, 0.4985, 0.4745, 0.4627$ for $h = 2, 3, 16, 20, 64, 2000$ (limit ≈ 0.4623): the $\frac{1}{2}$ barrier of Theorem 16(c) is *false* for value-aware rules from $h = 20$ on, and the whole class of in-gap rules is capped at ≈ 0.462 .

Proof (full text: [experiments/w31-lookahead/proof.md](#)). (a) The explored set of a step is $\{\psi \leq \psi(q)\}$, contained in $\{x' \leq e_j^{\text{new}}\}$, so searching beyond the safe clock keeps every search fresh (the sequential-search lemma of Theorem 11); the fresh part u of each increment depends on the past only through the strip's own state, so strip costs are independent and, for uniform π , i.i.d. with finite MGF ($u \leq_{\text{st}} \text{Exp}(C(1-2\varepsilon)\beta_h) + D_h$, $\beta_h = \varepsilon^{h-1}h$); collisions ($a_s > a$) need the global clock to advance less than D_h during the n_t intermediate steps of other strips, which has probability $\leq (eChD_h/n_t)^{n_t}$, and for uniform π only $O(h \ln k)$ steps have $n_t < \lceil e^2ChD_h \rceil \vee \lceil 3 \ln k \rceil$; Chernoff plus Markov. (b) The first value placed in a gap has uniform rank; the two sub-gaps carry independent uniform sub-patterns, and by scale covariance (relative margins, Φ homogeneous of degree -1) their costs are $\mathcal{V}_m/y, \mathcal{V}_{n-1-m}/(1-y)$ whatever the interleaving; the minimum of $x + A/y + B/(1-y)$ over a unit Poisson field has $\Pr(\min > s) = e^{-\text{area}(s)}$ with the sublevel set an interval by convexity; upper bounds propagate through the recursion by monotonicity of the minimum in the potentials. (c) Finite-horizon dynamic programme whose value function is $\sum_i V_{n_i}/G_i$ by induction. $\square$ We recomputed the recursion independently ($V_{64}/64^2 = 0.47453$ at $\varepsilon = 0$, 0.47615 at $\varepsilon = 0.05$, below the certified 0.47646 ; $W(1, 0) = 3$ by hand; the minimiser law checked by Monte Carlo). *Discussion.* The value-awareness of Theorem 16's rule was worth nothing provable; looking ahead in x inside the gap, with the exact mean-field cost-to-go, buys 0.038 of the 0.25 between $\frac{1}{2}$ and the identity's $\frac{1}{4}$, and Theorem 17(c) says that is all an in-gap rule can buy: the remaining gap to $\frac{1}{4}$ (and to the conjectured 0.22 for random π) needs information from outside the current gap — the half-strips of the other gaps, which the dynamic programme treats as irrelevant because the same rule will search them later — or a non-sequential construction. End-to-end simulation at $k = 2048$: 10/10

successes at $C = 0.50$, 0/10 at 0.46, mean cost within 1% of Ω_h/C plus the collision cost.

Theorem 19 (the cap for all sequential rules). Call a rule *sequential* if it places the positions of π in order, each point inside the gap between the already placed points of neighbouring values, chosen from a region A_p on which the process is conditionally Poisson given everything the rule has used (the hypothesis of the sequential-search lemma behind every threshold in this paper), and *non-anticipating* if it does not know the future values of π . Then for uniform $\pi \in S_k$ and Π_{Ck^2} , whatever else the rule looks at — other gaps, other strips, the whole fresh strip, arbitrary x -lookahead — its expected consumption of the x -clock is $\geq V_k/(Ck)$, with V_n the Bellman values of Theorem 17(c); so no such rule proves a constant below $\lim V_n/n^2 \approx 0.4623$, and a rule with a weak law for its consumption fails w.h.p. for every $C < V_k/k^2$. Knowing π in advance changes the cap by a factor $1 - 4 \cdot 10^{-5}$ (exact over all binary-search-tree shapes for $n \leq 11$). *Proof (experiments/w32-outofgap/proof.md).* Backward induction on the remaining values: given the state and the next value's gap i and rank m , the cost-to-go after the choice $q = (a + u, y)$ is $\geq u + \sum_{i' \neq i} V_{n_{i'}}/(CG_{i'}) + V_m/(C(y - y_L)) + V_{n_{i-1-m}}/(C(y_R - y))$; the bracket is $\varphi(q) \geq \min_{\Pi \cap A_p} \varphi \geq \min_{\Pi'} \varphi$ for $\Pi' \supseteq \Pi \cap A_p$ a Poisson process on the whole half-gap, whose expected minimum is $W(V_m, V_{n_{i-1-m}})/(CG_i)$ by scaling; averaging over the uniform rank and gap (non-anticipation) gives $\sum_i V_{n_i}/(CG_i)$. $\square$ *Consequence.* The route that produced Theorems 11, 16, 17 is exhausted at ≈ 0.46 for a typical pattern, against the numerical truth ≈ 0.22 : the missing factor 2 can only come from abandoning sequential commitment — a global (Hammersley/LIS-type) argument that chooses a copy after seeing the whole process — exactly the ingredient that Theorem 18's cross-strip lookahead supplies for rigid grids (there the cap is escaped because the rule commits a whole block at once).

Theorem 10 (block-grid patterns: an r -independent threshold). Let $k = rh$ and, for $\tau = (\tau_0, \dots, \tau_{h-1}) \in (S_r)^h$, let $\pi_\tau(ir + s) = \tau_i(s)h + i$ ($0 \leq i < h$, $0 \leq s < r$); the class $\mathcal{G}(r, h)$ of all such π_τ contains the tilted grid ($\tau_i = \text{id}$, i.e. the periodic word $(12 \dots r)^h$) and the whole family $\mathcal{F}(r, h, \varepsilon)$ of the previous paragraph. Let $N = Ck^2$ and let $c(C) = \sup_{\theta > 0} (\theta - \ln \mathbb{E} e^{\theta U})$, where (U, V) has density $Ce^{-C(u+v)^2/2}$ on the quadrant, so that $\mathbb{E}U = \sqrt{\pi/(8C)}$ and $c(C) > 0$ iff $C > \pi/8$ ($c(1) = 0.20$, $c(2) = 0.75$). Then for every $\pi \in \mathcal{G}(r, h)$

$$\Pr(\pi \notin \Pi_N) \leq e^{-c(C)k} + r e^{-c(C)h},$$

and by reflection also $\leq e^{-c(C)k} + h e^{-c(C)r}$. Consequently a uniformly random permutation of length $n = (\pi/8 + \varepsilon)k^2 \approx 0.393k^2$ contains every $\pi \in \mathcal{G}(r, h)$ with probability $1 - o(1)$ as $\min(r, h) \rightarrow \infty$, uniformly in π ; and at $n = k^2$ every block-grid pattern with $r \leq k/(6 \ln k)$ is contained with probability $\geq 1 - (r + 1)k^{-1.2}$.

Proof sketch (full proof: experiments/w11-strips/proof.md §4). Cut the values into r strips of h consecutive values and the positions into h slabs of r ; π_τ puts exactly one point in each (slab, strip) cell, visiting the strips of slab i in the order τ_i and, within a strip, ordered by slab. Complete each strip's points by an independent Poisson process outside the strip, so the r strip-processes are independent Poisson processes on the plane. Now embed greedily in position order: keep a global x -level a and a y -level b_j per strip, and at the cell $(i, \tau_i(s))$ take the point of the strip-process in the quadrant $\{x > a, y > b_j\}$ minimising $(x - a) + (y - b_j)$; advance both levels to it. By the Mecke formula the scaled increments $k(x - a), k(y - b_j)$ have, conditionally on the past, the fixed law (U, V) above (the explored triangles of earlier searches in the same strip are disjoint from the current quadrant, so the process there is still Poisson); hence all k increment pairs are i.i.d. On the event that the k x -increments sum to less than k and, in each strip, the h y -increments sum to less than h , every chosen point lies in its own cell (so it is a genuine point of Π_N) and the chosen points

form a copy of π_τ by construction; Chernoff bounds the complement. The rule $(x - a) + (y - b_j)$ is optimal among corner rules minimising a 1-homogeneous function (squares need $C > 9\pi/64$, quarter-discs $C > 4/\pi^2$). De-Poissonisation is the standard coupling of $\Pi_{(1-\delta)n}$ inside n uniform points. ■

Discussion. The greedy uses nothing about τ beyond bijectivity of each τ_i , so it is insensitive to exactly the structure (rows visiting strips in arbitrary orders) that made $\mathcal{F}$ defeat every chain-based hypothesis. Its threshold $\pi k^2/8$ is r -independent and below the *rigid-slab* threshold k^2 of the identity itself; the true threshold of these patterns is conjecturally $k^2/4$, so the loss is a factor $\pi/2$, the price of taking increments one cell at a time. What Theorem 10 does not give is a union bound: $|\mathcal{F}| = e^{\Theta(k \ln k)}$ while the failure probability has speed only $\min(r, h)$ in the rows (one bad row kills the greedy) — Theorem 4.4 of the proof file improves the row part to speed k , but the column part stays $re^{-c(C)h}$. So $\mathcal{F}$ is *not* a counterexample to Alon’s conjecture, and the obstruction it poses is to the method (per-pattern bounds summed over shift-chain classes), not to the statement. We checked the theorem numerically: the greedy, run with an assertion that the chosen points form π_τ , succeeds in 86% of trials at $r = h = 10$, $C = 1$ and never produced a non-copy. Exact threshold computations (a Pareto-front dynamic program for block-grid containment, validated against brute force on 7200 samples) show the fixed-strip threshold of the $r \times r$ tilted grid *decreasing* with r : $N_{1/2}/k^2 = 0.55, 0.50, 0.44, 0.41, 0.375, 0.36$ for $k = 4, 9, \dots, 49$, about 0.06 above the identity’s finite-size values, and in the free model every tilted grid with $k \leq 32$ is *easier* than the identity of the same length.

Theorem 11 (every pattern below k^2 : the reserve greedy). For every k and every $\pi \in S_k$, with $N = Ck^2$: (a) (rigid rows) $\Pr(\pi \not\subseteq \Pi_N) \leq \Pr(\text{Poisson}(Ck) \leq k - 1) \leq e^{-k(C-1-\ln C)}$, so the universal threshold is at most k^2 and a uniformly random permutation of length $(1+o(1))k^2 \ln k$ is a k -superpattern (the union bound with an explicit constant; He–Kwan’s threads in Poisson form). (b) Let C_{univ} be the threshold of the two-point reserve chain defined below; numerically $C_{\text{univ}} = 0.757$. For every $C > C_{\text{univ}}$ there is $\eta(C) > 0$ ($\eta(0.8) \geq 0.0014$, $\eta(1) \geq 0.035$, $\eta(1.2) \geq 0.096$) such that $\Pr(\pi \not\subseteq \Pi_{Ck^2}) \leq e^{-\eta(C)k+O(1)}$ for **every** $\pi \in S_k$; hence a random permutation of length $(0.757 + \varepsilon)k^2$ contains any fixed k -pattern w.h.p.

Proof sketch (full proof: [experiments/w19-general-greedy/proof.md](#) §2, §4B). (a) Cut the values into k rows of height $1/k$; embed the positions in order, each time taking the leftmost point of row $\pi(p)$ to the right of the current x -level. Rows are disjoint, so the scaled x -increments are i.i.d. $\text{Exp}(C)$ and the embedding fits iff $\text{Poisson}(Ck) \geq k$. (b) Cut the value sequence $\pi^{-1}(1), \pi^{-1}(2), \dots$ into maximal monotone runs and split them into pieces of length 2 and 3 (any two consecutive values are monotone, so this is always possible). Each piece is a strip of height $2/k$ or $3/k$ on which the positions are monotone. Embed in position order; for a strip with $m \geq 2$ points still to place, search *inside the strip* above the current level but keeping a **reserve** of β rows per future point, taking the point minimising $(x - a) + \lambda(y - b)$; the last point of a strip is the leftmost one in what remains. The strip’s own future depends on the past only through the height left in that strip, and different strips search independent (extended) Poisson processes, so the total x -cost is a sum of independent strip costs T_j with an explicit law; a copy of π exists as soon as $\sum_j T_j < k$, and Chernoff gives $e^{-\eta(C)k}$ whenever $\mathbb{E}T_j < h_j$ for both strip lengths. With $\beta = 0.4$, $\lambda = 1.1$: $\mathbb{E}T/2 = 1.0009$ at $C = 0.757$ and 0.947 at $C = 0.8$ for $h = 2$; $\mathbb{E}T/3 = 0.89$ at $C = 0.757$ for $h = 3$ (our independent simulation; W19’s quadrature agrees). ■

Discussion. Theorem 11(b) is the first threshold below k^2 valid for arbitrary patterns (He–Kwan give $20k^2$ for quasirandom patterns and $O(k^2 \log \log k)$ in general); the worst case is the alternating

pattern 214365..., whose value sequence consists of runs of length exactly 2. The gain over rigid rows comes from trading x -cost against the y -slack that rows waste. For structured π (long monotone value runs) one keeps the runs whole and the threshold decreases towards $\pi/8$. The constants are low-dimensional integrals, evaluated numerically and re-simulated independently; a certified evaluation is routine. **What this does not give:** the exponent is only ηk , and W19 proves (Props. 5.2–5.3 there) that every greedy of this kind — including two-phase repairs — fails with probability $\geq e^{-O(\min(k,h))}$, so no such procedure can reach the $e^{-(1+\varepsilon)k \ln k}$ per pattern that the union bound over S_k needs at $n = O(k^2)$; with these tools the union bound gives exactly the trivial $k^2 \ln k$. A proof of $n = O(k^2)$ must move all strips at once (a two-dimensional last-passage argument), which is where the next step lies.

What the standard concentration tools give. The classical toolbox (second moment, Janson’s inequality and its extensions, Talagrand, Azuma/McDiarmid, Kim–Vu, the local lemma) cannot supply the missing speed either, and for the main tools this is a theorem rather than a failure of ingenuity (`experiments/w22-probabilistic-method-review/review.md`). At $N = Ck^2$ every point of Π_N lies in $\approx \mu k/N$ copies of π ($\mu = \mathbb{E} \# \text{copies}$), so the pairs of copies sharing a single point already give $\Delta \geq (1 - o(1))\mu^2/C$ for *any* family of copy events, and the (extended) Janson exponent $\mu^2/2\Delta$ is at most $C/2$; the copy count is never approximately Poisson (exact second moments at $k = 4, 5$: $\text{Var} X/\mu^2 \asymp \sum_j c_j(\pi) C^{-j}/j!$, with clustering coefficients $c_j \approx j!$ for the identity and ≈ 1 for random-like patterns — a mechanism for the finite- k ordering observed above). Talagrand’s inequality on certifiable copy functionals gives exponent $\leq C/4$ (for the identity it yields the explicit speed- k bound $\Pr(\text{LIS} < k) \leq 2e^{-(2\sqrt{C}-1)^2 k/(8\sqrt{C})}$), martingale methods $\leq C/2$ (the deviation needed is $\Theta(\sqrt{N})$, the martingale scale), and the local lemma only proves positivity since all absence events are decreasing. The one global technique that is not capped is **certificate counting**: $\text{LIS}(\Pi_N) < k$ forces a Dilworth cover by $k - 1$ decreasing chains, and a union bound over the $(k - 1)^N$ assignments gives $\Pr(\text{LIS}(\Pi_N) < k) \leq (e/C)^N$ — speed N in three lines, without RSK. Containment of a general pattern has no such duality, but the residual classes do (copies of block-grid patterns are chains in a molecule poset, whose absence certificates are antichain covers), which is the natural next attempt.

Theorem 12 (the fixed-strip model of tilted grids: mean-field limit). Fix h and let $\pi_{r,h}$ be the $r \times h$ tilted grid ($k = rh$; the periodic word $(12 \dots r)^h$), embedded with *rigid* value strips of height $1/r$ (the model of Theorems 9–10; ordinary containment is the *free* model, which is easier). Call a per-strip rule *mean-field* if at each visit it takes, in a fresh half-strip above the strip’s current level, the point minimising $u + W_m(y)$ for bounded potentials W_m , and let $\mu_R(C)$ be its mean total x -cost per strip at intensity C . (a) If $\mu_R(C) < h$ then $\Pr(\pi_{r,h} \subseteq_{\text{fix}} \Pi_{Ck^2}) \rightarrow 1$ as $r \rightarrow \infty$; hence $\limsup_r C^{\text{fix}}(r, h) \leq C_{\text{mf}}(h)$, the root of the Bellman recursion $V_h(0; C) = h$, $V_m(v) = \int_0^\infty \exp(-C \int_v^h (\sigma - V_{m-1}(y))_+ dy) d\sigma$. (b) $C_{\text{mf}}(1) = 1$ (exact: for $h = 1$ the fixed model is word containment, constant 1), $C_{\text{mf}}(2) = 0.750$, $C_{\text{mf}}(h) = 1, 0.750, 0.655, 0.603, 0.570, \dots, 0.472$ ($h = 16$), and $\pi/8 \leq C_{\text{mf}}(h) \downarrow \pi/8$. (c) Every “fresh-quadrant” strategy (all corner rules, any lookahead inside the current sweep, all mean-field rules) pays $\mathbb{E}[u + v] \geq 2\sqrt{\pi/(8C)}$ per point, so none can beat $C = \pi/8$; the triangle rule of Theorem 10 is optimal in this class.

Proof idea (full proofs: `experiments/w20-hammersley-grid/proof.md`). Between two visits of a strip the shared x -clock advances by a sum of $r - 1$ increments of *other* strips, i.e. by $\Theta(r)$, while one visit explores a window of width $O(\ln r)$; so a strip has forgotten its previous exploration by its next visit. Formally one couples the process with an “artificial fill” of already-explored regions; the coupling fails with probability r^{-2} per step, and the strip costs are then i.i.d. and bounded, so Hoeffding finishes. The Bellman form is the finite-horizon dynamic programme of one strip against

fresh half-strips; the limit $\pi/8$ follows from (c) below and a coupling with the triangle rule. (c) is Hammersley's greedy bound applied to one quadrant. ■

Theorem 18 (cross-strip lookahead: block-grid patterns below $\pi/8$, numerically to $\frac{1}{4}$). In the fixed-strip model of Theorem 10 (rigid value strips of height $1/r$, $k = rh$), the barrier $\pi/8$ of Theorem 12(c) covers every rule whose current quadrant is still Poisson given its past; the only way to acquire information about it is to choose the point of strip s *after looking at the strips visited next*. Cut the visiting sequence into blocks of b or $b + 1$ consecutive visits by a stationary renewal partition independent of the process; in each block, among the fresh windows of its (distinct) strips, take the x -increasing path minimising $(x\text{-advance}) + \sum(y\text{-increments})$. Let K_b be the first-passage cost $\min\{x_b + \sum_{j \leq b} y_j : p_j \in \Psi_j, x_1 < \dots < x_b\}$ over b independent unit Poisson quadrants and $\gamma_b^{\text{mix}} = (\mathbb{E}K_b + \mathbb{E}K_{b+1})/(2b + 1)$. (a) For every $\pi_\tau \in \mathcal{G}(r, h)$ whose consecutive rows never revisit a strip within $b + 1$ visits (all tilted grids), $\min(r, h) \geq (\ln k)^3$ and $C > C_b^{\text{mix}} := (\gamma_b^{\text{mix}}/2)^2$: $\Pr(\pi_\tau \subseteq \Pi_{Ck^2}) \rightarrow 1$, uniformly in τ ; hence a uniform permutation of length $(1 + \varepsilon)C_b^{\text{mix}}k^2$ contains them w.h.p. (b) $C_b^{\text{mix}} = 0.369, 0.345, 0.321, 0.300, 0.284, 0.273, 0.266$ for $b = 1, 2, 4, 8, 16, 32, 64$ (Monte Carlo; $b = 1$ is the corner greedy, $\mathbb{E}K_1 = \sqrt{\pi/2}$ exactly), with $C_b := (\mathbb{E}K_b/2b)^2 \approx \frac{1}{4} + 0.105/\sqrt{b}$. (c) The globally optimal round (all r strips looked at) has cost per strip $\gamma_\infty = \lim \mathbb{E}K_n/n = 1$ *exactly* (Theorem 20 below; first found numerically as 1.000 ± 0.002 , $n \leq 8000$, by an exact suffix-minimum dynamic programme), i.e. the mean-field fixed-strip model with full lookahead has threshold $\gamma_\infty^2/4 = \frac{1}{4}$ — Alon's constant.

Proof (full text: [experiments/w34-grid-lookahead/proof.md](#)). Within a round the x -costs telescope, so with y -weight 1 the round is the first-passage problem (a); scaling $(x, y) \mapsto \sqrt{\lambda/C}$ shows $K_n(\lambda, C) \stackrel{d}{=} \sqrt{\lambda/C} K_n(1, 1)$, and differentiating in λ at 1 shows the minimising path spends exactly half its cost in x and half in y , so the per-visit mean x -advance and y -increment are both $\gamma/(2\sqrt{C})$ and the x -budget k and the y -budgets h per strip are met iff $C > \gamma^2/4$. A safe clock keeps every block's windows fresh (sequential-search lemma); offsets vanish w.h.p. because a strip is revisited only after $\geq r/(2b + 2)$ blocks; the stationary partition makes every visit sit at a uniform position of a size- b or $(b + 1)$ block, so each visit's mean x - and y -part is $\rho = \gamma_b^{\text{mix}}/(2\sqrt{C})$; Hoeffding on the block x -parts and Azuma plus renewal mixing on each strip's h independent y -increments, union over r strips ($h, r \geq (\ln k)^3$). □ We recomputed the constants with an independent first-passage code ($\mathbb{E}K_2 = 2.393 \pm 0.006$, $\mathbb{E}K_{64} = 66.2 \pm 0.2$, $\gamma_{1000} \approx 1.00$).

Discussion. This is the first rigorous constant below $\pi/8$ for block-grid patterns, and the exact diagnosis of what the barrier of Theorem 12(c) protects: not the corner search but the *absence of cross-strip lookahead*. W20's conjecture that mean-field rules are optimal at fixed h is false for large h ($C_2^{\text{mix}} = 0.345 < \pi/8$). The numerical identity $\gamma_\infty = 1$ (c) would mean that rigid strips lose *nothing* at first order for tilted grids — their fixed-strip threshold is $\frac{1}{4}$, the identity's value — which fits the picture that $\frac{1}{4}$ is the maximum over patterns (identity hardest) and that the sub- $\frac{1}{4}$ thresholds of $(12)^h$ at bounded r (Theorem 12 numerics) come from the non-mean-field, Hammersley-like regime. The Burke-type stationary solution of the backward recursion $G_s(a) = \min_{x_p > a} [y_p + G_{s+1}(x_p)]$ conjectured here as the route to $\gamma_\infty = 1$ is carried out in Theorem 20, which also replaces the Monte-Carlo constants of (b) by certified upper bounds.

Theorem 20 (Burke property of the cross-strip recursion; $\gamma_\infty = 1$; tilted grids at Alon's constant). Let $\Psi_1, \dots, \Psi_n$ be independent unit Poisson processes on $(0, \infty)^2$ and K_n the first-passage cost of Theorem 18. For $\rho > 0$ let G^ρ be the two-sided compound Poisson boundary with jump locations of rate ρ and i.i.d. $\text{Exp}(1/\rho)$ jumps (mean slope ρ^2), and $T_\Psi B(a) := \min\{y_p + B(x_p) : p \in \Psi, x_p > a\}$ the one-strip operator. (a) (Burke) $T_\Psi G^\rho$ is again a compound Poisson boundary

of rate ρ with i.i.d. $\text{Exp}(1/\rho)$ jumps, and $T_\Psi G^\rho(0) - G^\rho(0) \sim \text{Exp}(1/\rho)$; so the boundary laws G^ρ are exactly stationary under a strip step, with per-strip cost ρ . (b) $n - \sqrt{2n} \leq \mathbb{E}K_n \leq n + \sqrt{2n} + \frac{1}{2}$ for every $n \geq 1$, and $\mathbb{E}K_n \geq n$ for every n ; hence $\gamma_\infty = 1$ and $C_b, C_b^{\text{mix}} \geq \frac{1}{4}$ for every b — no block rule of Theorem 18 beats $\frac{1}{4}$. (c) $C_b^{\text{mix}} \leq (\frac{1}{2} + \frac{\sqrt{2b+\sqrt{2b+2}+1}}{2(2b+1)})^2 \downarrow \frac{1}{4}$ (0.350 at $b = 64$, 0.2572 at $b = 10^4$, 0.2507 at $b = 10^6$), all certified. (d) Consequently, for every $\varepsilon > 0$, every block-grid pattern $\pi_\tau \in \mathcal{G}(r, h)$ satisfying the hypothesis of Theorem 18(a) for some fixed $b = b(\varepsilon)$ — in particular every tilted grid $(12 \cdots r)^h$ — with $\min(r, h) \geq (\ln k)^3$ satisfies $\Pr(\pi_\tau \subseteq \Pi_{(1/4+\varepsilon)k^2}) \rightarrow 1$, and a uniform permutation of length $(\frac{1}{4} + \varepsilon)k^2$ contains it w.h.p.

Proof (full text: `experiments/w36-gamma-limit/proof.md`). (a) Write $D(a) := T_\Psi G^\rho(a) - G^\rho(a) \geq 0$; it depends only on the inputs to the right of a , so it is stationary. Conditioning on the boundary, $\Pr(D(0) > z) = \mathbb{E} \exp(-\int_0^\infty (z - G^\rho)^+)$, and splitting at the first jump gives the renewal equation $\varphi(z) = \frac{\rho}{\rho+z}(e^{-z/\rho} + \int_0^z \rho^{-1} e^{-j/\rho} \varphi(z-j) dj)$, solved by $e^{-z/\rho}$ (uniqueness by Grönwall) — this is where the pairing of rate ρ with mean ρ is forced. Read from right to left, D is a Markov jump process with holding rate $\rho + x$ that jumps up by $\text{Exp}(1/\rho)$ at the boundary's jumps and down to a uniform point of $(0, x)$ at the points of Ψ below it; $\pi(x) = \rho^{-1} e^{-x/\rho}$ satisfies detailed balance for exactly this pairing, so the process is reversible, and reversal exchanges the up-jumps (the boundary's) with the down-jump increments (those of the output), which is (a). (b) Iterating (a), $\mathbb{E}G_1^\rho(0) = n\rho$ for the n -strip system started from G^ρ ; the true boundary $B_0(x) = x$ is sandwiched, $G^\rho - Z' \leq B_0 \leq G^\rho + Z$ with $Z = \sup_{x \geq 0} (x - G^\rho(x))^+$, $Z' = \sup_{x \geq 0} (G^\rho(x) - x)^+$, and the operator is monotone and commutes with constants, so $n\rho - \mathbb{E}Z' \leq \mathbb{E}K_n \leq n\rho + \mathbb{E}Z$. Lundberg's exponent for the spectrally one-sided Lévy process $x - G^\rho(x)$ is $|\rho - \rho^{-1}|$, giving $\mathbb{E}Z \leq \rho/(\rho^2 - 1)$ ($\rho > 1$) and $\mathbb{E}Z' \leq \rho/(1 - \rho^2)$ ($\rho < 1$); take $\rho = 1 \pm 1/\sqrt{2n}$. The pointwise bound $\mathbb{E}K_n \geq n$ then follows from subadditivity ($\mathbb{E}K_{m+n} \leq \mathbb{E}K_m + \mathbb{E}K_n$ by concatenation at the optimal clock, which is independent of the later strips) and Fekete. (c) Insert (b) into the definitions. (d) Choose b with $C_b^{\text{mix}} < \frac{1}{4} + \varepsilon$ and apply Theorem 18(a); the hypothesis (no strip revisited within $b+1$ visits) holds for a tilted grid as soon as $r > b+1$, which $\min(r, h) \geq (\ln k)^3$ guarantees. $\square$

Discussion. (a) is the exact analogue of the sources-and-sinks stationarity of Hammersley's process (density ρ in, $1/\rho$ out), here in the form “jump rate $\rho \leftrightarrow$ jump mean ρ ”, and it pins the cross-strip constant to the LIS constant: a boundary of slope $\sigma = \rho^2$ costs $\sqrt{\sigma}$ per strip, and the free boundary x has $\sigma = 1$. So a rigid grid of value strips loses *nothing* at first order: the block-grid family reaches Alon's $\frac{1}{4}$ from above, the first non-monotone family to do so, and the whole gap $\pi/8 \rightarrow \frac{1}{4}$ of Theorems 10, 12, 18 is closed. Two caveats. The matching *lower* bound is not available: for these patterns the only lower bound in this note is the universal $0.1925k^2$ (LIS-type bounds do not apply — a tilted grid has $L(\pi_\tau) = r + h - 1 \approx 2\sqrt{k}$), and Theorem 12's numerics put $(12)^h$ at bounded r below $\frac{1}{4}$, so $\frac{1}{4}$ is an exact threshold only for the mean-field fixed-strip model, not (yet) for containment. And (b) is also a barrier: since $\mathbb{E}K_n \geq n$ always, block lookahead of any depth cannot go below $\frac{1}{4}$ — for a typical pattern, where the truth is ≈ 0.22 (the threshold numerics of §7), the remaining factor must again come from outside this family of rules, as in Theorem 19.

Discussion and numerics. Under the natural conjecture that the optimal (non-causal) fixed-strip embedding gains nothing over mean-field rules as $r \rightarrow \infty$ (true for $h = 1$; the exact Pareto-front dynamic programme confirms 0.58–0.61 vs 0.75 only up to $r = 16$ and is infeasible beyond), the fixed-strip constant in the iterated limit is exactly $\pi/8$: the corner greedy of Theorem 10 is asymptotically optimal *for rigid strips*, and the factor $\pi/2$ to the conjectured $1/4$ is the price of rigidity in the short direction, not of the greedy. The diagonal fixed model behaves differently — exact thresholds 0.55, 0.50, 0.44, 0.41, 0.375, 0.36, 0.335 for the $r \times r$ grid, $r = 2, \dots, 8$, already

below $\pi/8$ — so the diagonal and iterated limits differ. Most striking is the two-strip pattern $(12)^h = 1(h+1)2(h+2)\dots$: its exact fixed-strip threshold is 0.287, 0.268, 0.254 at $k = 64, 128, 256$ (i.i.d. colours give the same values), extrapolating to $\approx 0.22\text{--}0.23 < 1/4$, and in the free model it is 0.272 at $k = 64$ against 0.291 for the identity. Together with the random-pattern thresholds above this is fairly strong numerical evidence that **the identity is not the easiest pattern**: the constant $1/4$ in Alon’s conjecture is the *maximum* over patterns, attained by the monotone ones, and the true constant of $(12)^h$ is a new open quantity in $[1/8, 1/4]$ with no visible exactly-solvable structure (its dynamic programme is a two-dimensional Pareto front, not a Hammersley particle system). What remains open is any rigorous constant below $\pi/8$ for a grid with $\min(r, h) \rightarrow \infty$: by (c) it needs $\Theta(1)$ -precision coordination of the shared clock across strips, which destroys the renewal structure every proof so far relies on.

Symmetries and mixed directions. Containment is equivariant under the dihedral group (reverse, complement, inverse) and a uniform permutation is invariant, so any class covered w.h.p. can be replaced by the union of its eight images; the shift chains of π^c, π^r are those of π with LDS and LIS exchanged, and those of π^{-1} are the “horizontal” chains of π (pairs at horizontal distance Δ consistently ordered by value). The staircase count also extends to unions of ρ monotone chains of *either* direction (an extra factor $2^{\min(\rho, j)}$), so Theorem 5 holds for unions of increasing *or decreasing* runs on value intervals, or on position intervals, each run’s direction chosen freely.

Why chain-based hypotheses cannot finish the job. Three negative results delimit the thread method as it stands. (i) The entropy statement one would need for structured interleavings is false: for two runs whose value intervals are near-translates by Δ , the Δ -shift chain has length $\approx$ the run length *for every interleaving word* (the map $v \mapsto v - \Delta$ is monotone on both runs), so L_Δ carries no information about the word, and for $r \geq e^{75}$ there are at least e^{61k} patterns in $\mathcal{L}_r$ with only two mutually good shifts. (ii) More strongly, let $k = rh$ and $\pi_\tau(ir + s) = \tau_i(s)h + i$ for $\tau \in (S_r)^h$ with each τ_i having at least $(1 - \varepsilon)r$ fixed points (the He–Kwan tilted grid with perturbed rows). For $r, h \geq 20$, $\varepsilon \leq 1/16$, every dihedral image of every such π has $L_\Delta \geq k/16$ for *every* $\Delta \leq 0.7k$, so no three shifts can be chosen; and there are at least $e^{(1/32 - o(1))k \ln k}$ of them, too many for any bound of the form e^{-Dk} per pattern. Yet numerically the *actual* overlap of two threads on these patterns is $O(1)$ cells: threads at shift $\Delta = dh + e$ ride only when the follower’s element-lag $er + d$ is $\lesssim 3\sqrt{k}$. So what fails is the bound $|T_t \cap T_{t+\Delta}| \leq W_\Delta(z)$, which charges every chain, not the threads. One would hope for a *lag lemma*: a follower whose elements trail the leader’s by λ along a chain should have to overtake by $\approx \lambda$ columns before sharing a cell, at cost $e^{-\Omega(\lambda^2/k)}$, so that only $O(\sqrt{k} \log k)$ shifts are harmful for any pattern — which would give $n = O(k^2)$ for all of S_k . **This is false.** (iii) Let π be the tilted grid $\pi(ir + s) = sh + i$ on $k = rh$ elements, $m = Ck$ columns, and let E_t be the failure event of the thread at shift t . For every shift $e \leq h/4$ (element lag $\lambda = er \in \{r, 2r, \dots, k/4\}$, all $\geq \sqrt{k}$ when $r = h$) and $C \geq 3, k \geq 6$,

$$\Pr(E_0 \cap E_e) \geq 2(144C^2)^{-\lambda} \Pr(E_0),$$

and symmetrically for $e \geq 3h/4$; hence for $C \geq 8$ every such shift has $\Pr(E_0 \cap E_e) \geq \Pr(E_0)^2 e^{\Omega(k)}$. The proof prescribes $4\lambda - 1$ cells (the leader finds its first 2λ elements at columns $0, \dots, 2\lambda - 1$; the follower spends columns $2b, 2b + 1$ on element $b < \lambda$), after which the two threads sit in the same row at column 2λ and read identical cells forever (for $a \geq 2\lambda$, $\pi(a - \lambda) + e = \pi(a)$); the remaining cost is a binomial shift inequality $f(m - j, k - 1 - j) \geq ((k - 1 - j)/m)^j f(m, k - 1)$ for $\Pr(\text{Bin}(m, \frac{1}{2}) \leq k - 1) = f(m, k - 1)$. The loss is *linear* in the lag, not Gaussian; the $\sqrt{k}$ threshold seen in the simulations is an artefact of the unconditioned regime (both threads at speed 2 with a diffusive offset), whereas conditioned on failure a thread’s runs are geometric at

scale m/k and coalescence is cheap. The mechanism uses no long zero-run, so it survives every global event so far considered $(\mathcal{A}_{L,s}, \mathcal{B}_{L,s,\rho})$. (An exact column-by-column Markov computation of $\Pr(E_0 \cap E_\Delta)$ confirms the rate: 2.0–2.3 nats per unit lag at $C = 8$, independent of $k \in \{100, 400\}$, and $\approx \ln C$ across $C \in [3, 24]$; the best shift $\Delta = h/2$ is worth 45% of an independent thread. For a *random* π at $C = 8$ every shift has $\Pr(E_0 \cap E_\Delta)/\Pr(E_0)^2 = e^{140 \pm 10}$ at $k = 100$ — the run-concentration effect that the global events exist to remove, so some global event is indispensable in any thread argument.) Consequently (heuristically beyond two threads, since a follower rides whichever earlier thread has the smallest lag and the lags to nearest earlier shifts sum to $\leq (k/2)(1 + \log_2 \ell)$) ℓ threads on the tilted grid yield an exponent at most $Ck \ln 2 + O(k \ln C \log \ell)$, while the family $\mathcal{F}(\sqrt{k}, \sqrt{k}, 1/16)$ needs $\geq (1/32)k \ln k$ per pattern: **the thread framework with polylogarithmically many threads at $m = Ck$ cannot prove $n = O(k^2)$** , whatever lag lemma one proves — the same $k \log \log k$ barrier He and Kwan hit. Details: `experiments/w14-entropy/proof.md`, `experiments/w15-symmetry/proof.md`, `experiments/w18-lag/proof.md`.

What remains, precisely. The $\log \log k$ is **not** removed. With $\ell \approx \ln k$ threads the chains must be shorter than $k/(\ell \ln k)$, so He–Kwan’s structured decomposition must tolerate $\gtrsim \ln^3 k$ exceptional shifts, whose count is $e^{\Theta(k \log \log k)}$ — tight for unions of $r \approx \ln^2 k - \ln^3 k$ increasing runs on value intervals. After Theorem 5 and the symmetry extension, the class still unhandled is exactly the patterns with long shift chains for most shifts in every dihedral image, of which the perturbed tilted grids above are an $e^{\Theta(k \ln k)}$ -size example; by (iii) these *are* hard for the threads themselves, not only for the chain bound — yet by Theorem 10 each of them individually has threshold $\leq 0.4k^2$, so the difficulty is purely one of aggregation. For these, amplification fails (chains of length $\approx 1.3k/\sqrt{r}$ are too long) and the union bound fails (it would need $\approx \log r$ independent threads). A per-pattern bound $\Pr(\pi \not\subseteq \sigma) \leq e^{-k \log r}$ at $n = Ck^2$ for exactly these π would complete the proof of $n = O(k^2)$. We also showed that per-pattern chain accounting alone cannot give it: conditioned on a thread’s failure its zero-runs are geometric at scale m/k , so a heavy chain costs only $e^{-\Theta(k)}$ and every later thread rides for free — some global event is unavoidable, and its price is the union bound over the objects it controls. Since (iii) shows that the lag mechanism cannot be the way out, a proof of $n = O(k^2)$ for the perturbed tilted grids must leave the thread framework: either $\Theta(\log k)$ threads at $m = Ck$ with genuinely independent exposures (e.g. threads in the eight dihedral images, or threads restricted to disjoint column blocks), or the point-process methods of the previous section (canonical copies, comparison with the identity), which handle all patterns at once.

8 Verification and reproducibility

- **Theorem A** is a short, self-contained proof (§2). The constant λ_A is the root of an explicit one-variable convex minimization; it was computed independently twice (SciPy and mpmath at 30 digits) and the finite- k inequality (A) was evaluated directly at $k = 10^5$.
- **Theorems 2–4** ship with witnesses that any reader can re-check: a pure-Python brute force for Theorem 2 (`experiments/witnesses/check_witness.py`: all $\binom{23}{7} = 245,157$, $\binom{30}{8} = 5,852,925$ and $\binom{24}{7} = 346,104$ subsets are rank-standardised and counted; self-tests on `sp(3)–sp(6)` witnesses and on damaged witnesses; SHA-256 of `witnesses.txt`: `42b52a62...e7be3e4`), a C depth-first checker with different standardisation and enumeration code, and the Lean kernel-checked checker of (i) below, the SAT scripts and UNSAT logs for Theorem 3, the C checker for Theorem 4. All code is in `experiments/`; the witnesses, two independent checkers and their SHA-256 hashes are in `experiments/witnesses/`.
- **Lean 4**. A Lean 4 project (`formal-verification/lean/`, Lean and Mathlib v4.33.1) builds completely (`lake build`, 8715 jobs) and prints an axiom audit. It contains:

- (i) *Certificates.* Definitions of order-isomorphism, pattern containment and `IsSuperpattern k`; a DFS checker with the kernel-checked soundness theorem `checker_sound : checker k = true → IsSuperpattern k` (axioms `propext`, `Classical.choice`, `Quot.sound` only); and the theorems `IsSuperpattern 7` (length 23) and `IsSuperpattern 8` for the witnesses of lengths 30, 31 and 32, discharged by `native_decide` (about 5 s for $k = 7$ and 67 s CPU for $k = 8$, $n = 30$). `native_decide` adds one per-theorem axiom recording trust in the Lean compiler; the $k = 3, 4$ examples are proved by the kernel’s `decide` alone, while kernel `decide` for $k \geq 5$ did not finish (the kernel evaluates array updates as list operations).
- (ii) *Theorem A.* `theoremA` — for `.Nodup`, $k = 2m + 1$ and $0 < x < 1$, `patCount k (x^(n+1)) ^ 1 (x/(1-x))^(k+1) (1-x^k)^m` — is **fully proved**: `lake build` reports that `theoremA`, `lemma1_concrete` and `patCount_le_sum_W` depend only on `propext`, `Classical.choice` and `Quot.sound`. The formalization comprises the finite form of Lemma 2 (`pair_sum_le`), the tilt factorization (`tilt_bound`), the gap bijection (`compSum_gaps`, `gaps_mem_box`, `gaps_inj0n`), the abstract fibre-counting form of Lemma 1, the CKS injectivity in counting form (`card_patterns_over_le`) and the extension count (`fiber_card_ge`).
- (iii) *Constant.* `constant_certificate`, i.e. $1.0003 \cdot 7.37/e^2 - \log 7.37 + 1 + \frac{1}{2} \log(1 - e^{-7.37}) < 0$, sorry-free, from Mathlib’s rational bounds on e and $\log 2$, the Taylor bound for $\log(1 - 63/800)$, and `exp_bound` for $e^{0.37}$ (hence $e^{7.37} \leq 1588$). The true value is $\approx -1.25 \times 10^{-5}$, so the bounds had to be tight.
- (iv) *Proposition 15, block splitting, Erdős–Szekeres.* `Witness.lean` formalises Proposition 15 in a finite weighted probability space (`FinProb`): `sq_mean_le` ($\mu^2 \leq \mathbb{E} M^2 \Pr(M > 0)$), `E_sq_eq_sum`, `mean_le_max_cond` ($R \leq \max_{\pi} \Lambda_{\pi}$), `witness_reduction_max` ($\mu \leq \max_w \mathbb{E}[M \mid W = w] \Pr(M > 0)$) and its instance `witness_reduction_uniform` for the uniform law on S_n with M the number of missing k -patterns (with $M > 0 \iff \neg \text{IsSuperpattern}$). `BlockSplit.lean` proves the combinatorial core of block splitting, `blockStd_avoids` (containment is hereditary under restriction to a position block and standardisation; `contains_ranks_iff`), and `ErdosSzekeres.lean` connects Mathlib’s Erdős–Szekeres to the project’s containment (`not_avoid_id_rev`: for $n > (k - 1)^2$ the identity and its reverse are never both missing). All sorry-free, standard axioms only (audit reproduced by us). Left out, not sorried: the probabilistic block-splitting inequality $p_{\pi}(N) \leq p_{\pi}(M)^{\lfloor N/M \rfloor}$ (equidistribution of block standardisations). Reproduce with `cd formal-verification/lean && lake exe cache get && lake build; formal-verification/lean/LOG.md` records versions, timings and the `#print` axioms output.

9 Novelty audit

Before calling a result new we searched under the neighbouring terminology (superpatterns; universal and k -universal permutations; universal/complete sequences for patterns; dense packing of patterns; zigzag and circular constructions; OEIS A342474), read the papers citing [2] and [3] up to August 2026, and compared every witness with the known ones under the dihedral symmetries (reverse, complement, inverse): the length-23 and length-30 witnesses are not tie-broken words over $[k + 1]$ in any symmetric image (minimal monotone-row alphabets 9–10 and 13), so they are not disguised Engen–Vatter or Arnarson-type constructions, whereas Arnarson’s 17 *is* a tie-broken three-block word (§4). The unpublished 15/32 upper-bound claim of Hunter is recorded but not

used. Details: `memory/priority-check.md`. What has not yet been done is to ask specialists in permutation patterns about prior art or a definitional mismatch; until then the results should be read as a preprint-stage claim.

10 Discovery method and AI disclosure

Declaration of AI-assisted research. This work was carried out on 2026-08-29 with Claude Code (model Claude Fable 5, `claude-fable-5`) as the discovery engine: a coordinating session read the literature, ranked attack lines, and spawned parallel workstreams (`experiments/w1-w18`), each of which wrote code, ran searches, and drafted proofs. The human author set the goals, chose which directions to pursue, and questioned or rejected claims; the coordinating session independently re-derived every theorem produced by a workstream, recomputed every numerical certificate with separately written code, and brute-forced combinatorial lemmas per class before anything entered this paper — one lemma (an early draft of the value-slot refinement in §2) was false and was caught in this way, and the entropy hypothesis and the lag lemma of §7 were shown to be false rather than assumed. The simulated-annealing witnesses were found with random seeds recorded in `experiments/w1-search/logs/`; SAT results use PySAT/CaDiCaL with the encodings in `experiments/w4-alphabet-rosary/` and `experiments/w6-exact-sp7/`. All workstream briefs, logs, dead ends, code and outputs are archived in the repository; the LLM is not an author, and the human author takes full responsibility for every claim, citation and program. Every bibliographic entry was checked against arXiv/DOI metadata.

11 Open problems

1. Is $\text{sp}(7) = 23$? A proof would need exhaustive search over S_{22} modulo the dihedral symmetries — the $k = 6$, $n = 16$ case was done by Pantone; $n = 22$ is currently out of reach: a SAT/CEGAR formulation (order-matrix variables plus per-pattern prefix chains) re-proves $\text{sp}(4) = 9$ instantly — the 14 layered and co-layered 4-patterns alone form a minimal unsatisfiable core at length 8 — and $\text{sp}(5) = 13$ in about 20 s, but the cost grows by more than two orders of magnitude from (5, 12) to (6, 16), which did not finish. Note also that the 226 layered and near-monotone 7-patterns *are* simultaneously realizable at length 22, so any unsatisfiable core there must be larger.
2. Certify $\lambda = 1.000437$ (two-parity rate) rigorously.
3. Improve Theorem C' (1.00734): both parities (as in Theorem B) and keeping $\prod f(b_i)$ on the bad event should push further towards the heuristic ceiling 1.0117 (the remaining loss is now the good-factor slack, not the tail bound); formalize Theorems C/C' in Lean.
4. $f(6; 7)$; $r(6)$, $r(7)$; Gupta's conjecture for odd $n \geq 13$.

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